

Master Thesis

The Impact of Int8 Embedding Quantization on High-Precision Information Extraction from Dutch Procurement Documents

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Abstract

This thesis investigates the impact of 8-bit scalar quantization (Int8) on the dense semantic retrieval model `e5-multilingual-large` when extracting high-precision information from specialized Dutch procurement documents. Utilizing a real-world validation dataset of 16,819 Dutch tender snippets and 172 queries, the study balances engineering efficiency against semantic fidelity by comparing Dutch retrieval performance with a parallel, machine-translated English baseline.

At the macro level, evaluation metrics show notable resilience. Int8 quantization achieves a 75% index memory reduction with a negligible performance delta, maintaining mean NDCG@10 scores of 0.5969 (Float) versus 0.5987 (Int8) for the Dutch dataset. However, granular query-level error tracking exposes a micro-level paradox: while quantization regularly acts as a beneficial low-pass semantic regularizer on generic text by smoothing away low-frequency syntactic noise, it triggers severe localized degradations when faced with non-textual structural elements and alphanumeric criteria. To isolate this mechanism, a controlled synthetic stress test shows that these layout artifacts (e.g., emojis, currency symbols) act as high-magnitude vector outliers. This induces severe 'axis-stretching', shifting an average of 90.60% of embedding dimensions across integer boundaries and collapsing rank stability ($\tau = 0.8112$) within affected sub-spaces.

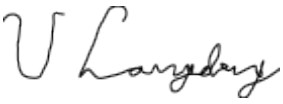
Furthermore, cross-lingual analysis demonstrates that this micro-level volatility manifests asynchronously across languages due to language-specific syntax mapping. Compact Dutch compound words face intense boundary clipping under quantization, whereas distributed English multi-word phrasings offer a structural buffer against severe document reordering. Ultimately, this research demonstrates that while Int8 quantization is exceptionally viable for production-scale pipelines, its application in high-stakes legal-administrative domains requires targeted validation to mitigate layout-induced and language-specific geometric drifts.

Declaration of Authorship

I, Vera Langeberg, declare that this thesis, titled *The Impact of Int8 Embedding Quantization on High-Precision Information Extraction from Dutch Procurement Documents* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 21 – 06 – 2026

Signed: 

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Finally, I would like to thank Luís de Passos Morgada da Costa for serving as the second reader of this thesis and for his time and effort in reviewing this work.

Additionally, generative artificial intelligence tools were utilized during the preparation of this thesis. Gemini 2.5 Flash was employed strictly for grammatical polishing, spelling corrections, and improving text readability, while Claude was used as an assistant for code debugging and optimization. All primary research, experimental design, underlying logic, and data analysis remain entirely the independent work of the author.

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Chapter 1

Introduction

Modern public and private procurement networks handle a massive influx of tender specifications, legal boundaries, and administrative mandates daily. In the European Union alone, public procurement is a massive economic driver, accounting for roughly 14% of the EU's Gross Discounted Product (GDP) (European Commission, 2025), which equates to over €2 trillion in annual spending, requiring strict adherence to rigid legal, technical, and compliance-driven guidelines. Extracting high-precision information from these unstructured, multi-page procurement documents is a highly labor-intensive task, yet prone to extreme human error. A single overlooked technical parameter, an unmapped ISO certification requirement, or a miscalculated budget ceiling can result in immediate bidder disqualification, severe financial losses, or major legal disputes.

To streamline this workflow, modern Information Retrieval (IR) and Retrieval-Augmented Generation (RAG) frameworks have largely transitioned from brittle, keyword-based lexical matching systems like BM25 (Robertson and Zaragoza, 2009) toward dense semantic retrieval pipelines powered by deep transformer-based bi-encoders (Karpukhin et al., 2020). By projecting text snippets into continuous, high-dimensional vector spaces, models such as `e5-multilingual-large` inherently support multiple languages. This allows systems to capture contextual nuances, implicit intent, and legal synonyms directly from the source text without the need of external translation steps (Wang et al., 2024).

1.1 The Efficiency-Precision Paradox

Despite the remarkable retrieval quality offered by state-of-the-art dense embedding models (Wang et al., 2024), implementing semantic search at a production scale presents severe computational bottlenecks. Storing vast document corpora as full-precision (`float32`) 1024-dimensional vectors incurs massive Random Access Memory (RAM) footprints and high infrastructure costs, which rapidly scale past commercial sustainability. To circumvent these physical hardware limits, exact search methods are bypassed in favor of Approximate Nearest Neighbor (ANN) search index structures like Hierarchical Navigable Small World (HNSW) graphs to achieve logarithmic complexity during search operations (Malkov and Yashunin, 2018). In modern database frameworks like Elasticsearch, these structures are typically coupled with aggressive vector compression techniques, chief among which is 8-bit scalar quantization (`Int8`).

By mapping continuous mathematical ranges into 256 discrete integer intervals, `Int8` quantization reduces the memory footprint of a vector index by 75% and allows similarity search to utilize highly efficient integer arithmetic natively supported by modern database

backends. However, this engineering convenience introduces a profound mathematical trade-off: rounding boundaries inevitably collapse subtle, fine-grained semantic differences into identical integer bins. While standard academic benchmarking datasets (e.g., MTEB, MIRACL) suggest that this compression results in negligible macro-level drops in accuracy (Wang et al., 2024), there remains a critical research gap regarding how compressed vector spaces handle the dense, hyper-specific administrative and legal structures of out-of-distribution domains.

1.2 Domain Adaptation and Query-Level Robustness

Beyond broad computational metrics, the precision-efficiency trade-off exposes a significant vulnerability when embedding models are applied to specialized professional fields without domain adaptation (Wei et al., 2025). When a multilingual model is quantized, its internal mathematical boundaries are rewritten. In standard datasets, this rounding noise goes unnoticed. However, in technical domains such as Dutch procurement, the vocabulary layout is highly unique, relying on complex single-token compound nouns (e.g., *gunningscriteria* [award criteria], *overmachtsbepalingen* [Provisions for unforeseen circumstances], *gestanddoeningtermijn* [tender validity period]) and strict alphanumeric codes.

According to Proskurina et al. (2023), transformer models subjected to heavy compression can experience an unexpected amplification of internal biases and a distortion of model confidence. This phenomena is in their paper called "the other side of compression" (Proskurina et al., 2023). Furthermore, Proskurina et al. (2024) demonstrate that post-training quantization does not affect a system uniformly. Instead, it disproportionately degrades performance on edge cases where the full-precision model already exhibited lower confidence levels in the first place (Proskurina et al., 2024). In practical procurement systems, this means that while a compressed system might retain stable performance for common corporate text, it may experience localized, unpredictable failures on hyper-specific legal specifications or complex technical criteria.

Crucially, this vulnerability extends beyond standard text to structural and formatting elements. Unique layout features such as table structures or non-standard characters like warning emojis act as high-magnitude vector outliers. As demonstrated by Dettmers et al. (2022), the presence of such outliers artificially stretches the quantization scale, exhausting the limited integer dynamic range and flattening adjacent, fine-grained semantic terms into identical bins. Investigating these micro-level query variations is therefore essential to determine whether automated procurement pipelines systematically miss critical technical constraints due to hidden compression artifacts.

1.3 Research Scope and Objectives

This thesis investigates how Int8 semantic compression interacts with complex document formatting and specific terminology in Dutch procurement. Instead of evaluating the model based solely on broad, average accuracy scores, this research examines individual search queries to try to pinpoint how and why specific layout features and terms cause the retrieval process to fail. By using a proprietary, real-world validation dataset of 16,819 Dutch tender snippets and 172 targeted queries, this research investigates how quantization interacts with complex domain terminology and document layouts. Given the specialized nature and potential sociolinguistic implications of using automated translation on legal-administrative

procurement documents, a comprehensive description of the corpus curation, speaker demographics, and data properties is documented in the full Data Statement provided in Appendix A - Data statement. Furthermore, this research integrates a parallel, cross-lingual track using an automated English translation proxy to isolate structural architectural behaviors from language-specific vocabulary layouts.

The primary research question guiding this investigation is formulated as follows:

> *To what extent does Int8 embedding quantization in the e5-multilingual-large model impact the retrieval performance, rank stability, and query-level error patterns for high-precision information extraction from specialized Dutch procurement documents compared to a parallel, English-translated control track?*

To systematically resolve this core question, this study pursues four key research objectives:

- **Macro-Level Performance Resilience:** To evaluate the aggregate impact of `Int8` quantization on standard retrieval metrics ($NDCG@10$ and MRR) to determine if compression preserves global retrieval capabilities across both Dutch and English datasets.
- **Micro-Level Query Pattern Analysis:** To examine individual query variations and identify why specific administrative themes experience performance regressions due to layout-induced vector outliers, or conversely, achieve performance gains through quantization-induced semantic regularization.
- **Cross-Lingual Comparative Analysis:** To compare retrieval performance, rank stability, and query-level quantization effects between the original Dutch procurement corpus and an English-translated version of the same dataset, identifying whether observed quantization patterns generalize across language representations.
- **Controlled Structural Validation:** To isolate and empirically validate the geometric mechanics of non-textual layout features (emojis, formatting markers, and currency keys) on linear discretization intervals using a controlled, synthetic injection framework.

1.4 Thesis Outline

The remainder of this thesis is structured as follows. Chapter 2 provides a theoretical overview of dense vector spaces, scalar quantization mechanics, and multilingual model properties. Chapter 3 describes the data preparation pipeline, the Dutch-to-English translation proxy, the technical implementation of the evaluation framework and the description of the synthetic dataset "stress test". Chapter 4 reports the empirical findings from both language pipelines and the synthetic dataset, highlighting macro stability and localized rank shifts. It also discusses the results of the special character stress test, showing a blurring effect. Chapter 5 offers a deeper insights, exploring the structural regularizing effects of quantization, language-specific subword fragmentation dynamics, and the axis stretching theory. Finally, Chapter 6 summarizes key findings and outlines future research directions.

Chapter 2

Background

This chapter establishes the theoretical and technical foundation for the research. It details the mechanics of dense semantic retrieval and graph-based indexing (2.1), the mathematical principles of vector quantization (2.2), and the performance of multilingual embedding models (2.3). Finally, it outlines the domain-specific challenges of Dutch procurement documentation (2.4) before defining the core research gap addressed by this study (2.5)

2.1 Semantic Retrieval and Graph-Based Indexing

Traditional text retrieval systems relied primarily on lexical matching frameworks like BM25 (Best Match 25). BM25 is a probabilistic ranking function that evaluates keyword relevance by calculating term frequency and inverse document frequency within an inverted index structure (Robertson and Zaragoza, 2009). While highly efficient, lexical approaches suffer from vocabulary mismatch; they evaluate exact string matches rather than meaning. Consequently, a system using lexical matching fails to connect synonymous terms or capture contextual nuances, meaning a search for "car" would fail to retrieve a document containing only "automobile".

To overcome these linguistic limitations, modern Information Retrieval (IR) leverages semantic retrieval architectures that use dense vector spaces. Instead of matching exact words, these frameworks project textual data into a continuous, high-dimensional manifold where text snippets are mapped via dense semantic embeddings based on their underlying meaning rather than their literal spelling (Karpukhin et al., 2020).

This vector space is constructed using a Bi-Encoder architecture. A Bi-Encoder uses a deep transformer-based language model to encode queries and documents independently. Unlike Cross-Encoders, which process a query and a document simultaneously to score them (a method that is highly accurate but computationally too slow for large databases), a Bi-Encoder processes text in isolation. One branch of the architecture encodes the user query, while a separate, parallel branch encodes the database documents.

In this study, the pretrained `e5-multilingual-large` model serves as the primary bi-encoder framework. When raw text, such as a Dutch procurement snippet, is fed into this model, it outputs a dense vector: an array of 1,024 continuous floating-point numbers (`float32`) representing the text's latent semantic coordinates. Because both queries and documents are mapped into this identical 1,024-dimensional space, their semantic similarity can be rapidly calculated using basic geometric operations like cosine distance or inner product calculations.

Once these high-dimensional vector representations are generated, the system must

search through them to find the closest matches. However, performing an exact k-Nearest Neighbor (k-NN) search requires comparing a query vector against every single document vector in the database, resulting in a linear time complexity of $\mathcal{O}(n)$. As database volume scales, this brute-force comparison becomes a major computational bottleneck. To achieve

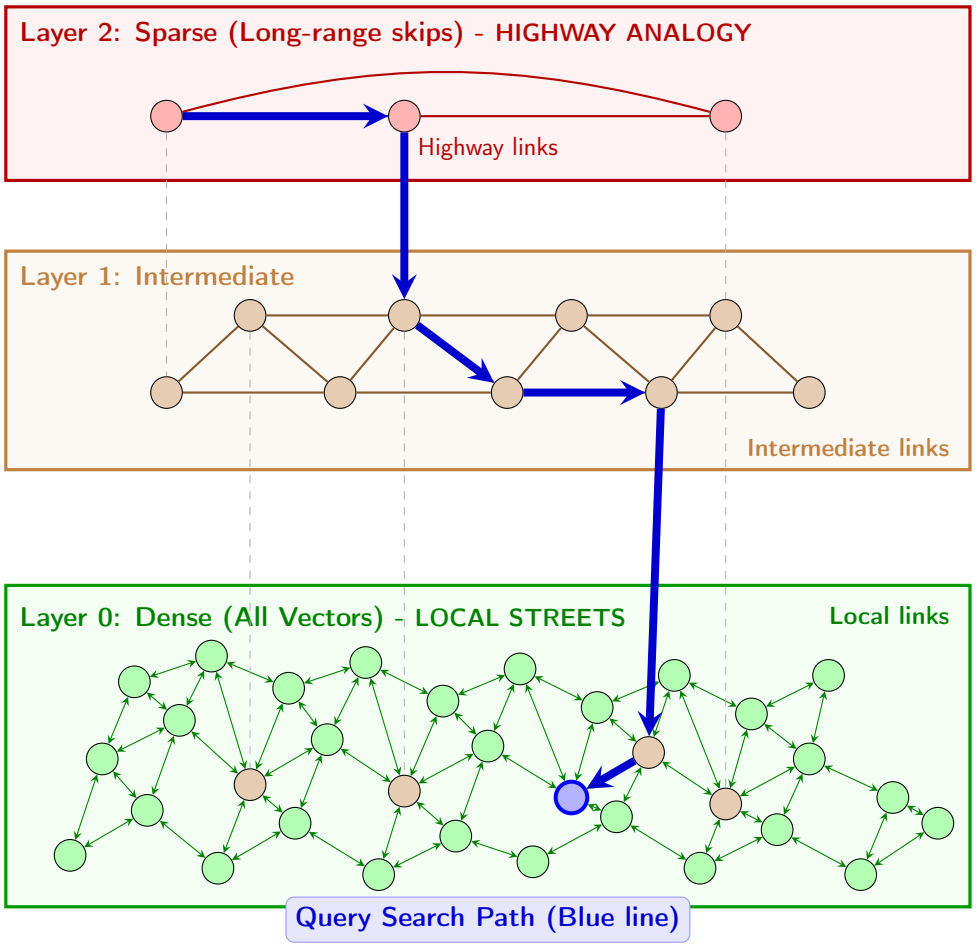


Figure 2.1: Hierarchical Navigable Small World (HNSW) search execution pattern across varying layer densities. The search query (blue trajectory) begins at the sparse top layer (Layer 2, representing long-range highway links), scales downward to find entry points in the intermediate tier (Layer 1), and concludes with fine-grained local routing in the dense base layer (Layer 0). Dashed gray lines denote vertical projections of identical vector nodes across layers, and the targeted destination vector is highlighted in blue.

production-grade search speeds, systems bypass exact matching in favor of Approximate Nearest Neighbor (ANN) search algorithms, specifically the Hierarchical Navigable Small World (HNSW) graph approach (Malkov and Yashunin, 2018). The HNSW algorithm organizes the high-dimensional vector space into a multi-layered graph structure, conceptually similar to a skip-list or a highway transit network.

As illustrated in 2.1, the top layers of an HNSW index consist of sparse graphs with long-range links between vectors, allowing the search algorithm to make massive "skips" across the data space to quickly find the general neighborhood of interest. As the algorithm descends into the lower layers, the graph links become increasingly dense and localized. At the bottom layer, which contains every vector in the dataset, the algorithm performs short-

range, fine-grained routing to pinpoint the closest semantic matches. This multi-layered structure reduces the search complexity from linear to logarithmic ($\mathcal{O}(\log n)$) (Malkov and Yashunin, 2018).

In a production pipeline, these distinct components are glued together within a unified database system. This research uses the Elasticsearch framework, which natively supports the HNSW indexing algorithm. The end-to-end architecture operates as follows: during the ingestion phase, Dutch tender documents are passed through the `e5-multilingual-large` model to generate 1,024-dimensional `float32` vectors, which Elasticsearch then indexes into an HNSW graph structure. At query time, an incoming search query undergoes the exact same vector conversion, and Elasticsearch routes the resulting query vector through the HNSW graph layers to retrieve the top semantic matches in milliseconds. However, while HNSW provides exceptional search velocities, storing millions of 1,024-dimensional vectors in their native `float32` precision creates a massive memory footprint, serving as the core motivation for exploring vector quantization techniques.

2.2 Principles of Vector Quantization

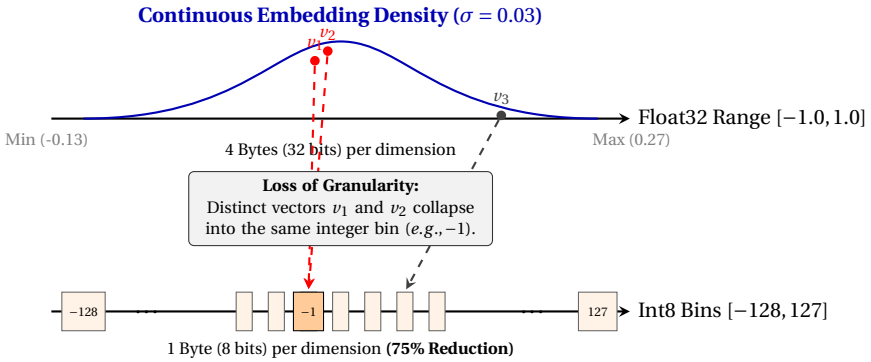


Figure 2.2: Visualization of Scalar Quantization. The continuous float32 distribution is discretized into 256 equal-width intervals. Highly similar technical parameters (v_1 and v_2) lose their fine-grained boundaries as they collapse into a single shared integer bucket.

Standard embeddings are generated in `float32` precision, where each dimension occupies 32 bits (4 bytes). For a model like `e5-multilingual-large`, which produces 1024-dimensional vectors, a single document requires 4,096 bytes of memory. Int8 quantization scales and shifts these floating-point values into an 8-bit integer range (-128 to 127). This reduces the memory footprint to 1 byte per dimension, yielding a 75% reduction in memory size.

In the context of this study, this is achieved through a scalar binning technique. This process involves defining a range $[min, max]$ derived from the distribution of the embedding values and dividing that range into 256 discrete bins. Each continuous `float32` value is then mapped to the nearest bin index, as illustrated in Figure 2.2.

Beyond storage efficiency, this transition from a continuous to a discrete vector space introduces complex effects on model behavior. Proskurina et al. (2024) note that this loss of granularity, where unique semantic nuances are compressed into the same integer bin, often results in a measurable decrease in the model's confidence regarding true labels or correct semantic matches. Furthermore, their findings suggest that quantization dispro-

portionately affects samples where the original full-precision model already exhibited lower confidence.

In the context of Dutch procurement, this implies that borderline retrieval cases, where a tender requirement and a candidate document share a complex or subtle semantic link, may be the first to fail under Int8 compression. As the binning process collapses the distance between highly similar but distinct technical terms, the model's ability to distinguish between "nearly correct" and "precisely correct" specifications may be compromised.

2.3 Performance of Multilingual Embedding Models

The `e5-multilingual-large` model is a state-of-the-art transformer-based encoder whose retrieval performance is derived from a specialized, multi-stage training pipeline (Wang et al., 2024). Rather than generating representations from scratch, the framework initializes from a pretrained `xlm-roberta-large` backbone, inheritely capturing foundational cross-lingual syntax across approximately 100 languages.

Following initialization, the training architecture operates via a two-stage paradigm:

1. **Weakly-Supervised Contrastive Pre-training:** The model is continually trained on a colossal corpus of roughly one billion automatically harvested multilingual text pairs (e.g., Reddit comment threads, translation bitexts, and Wikipedia hierarchies). Using a contrastive learning objective, the model minimizes the distance between semantically related "positive" text pairs within the 1,024-dimensional space while maximizing the geometric distance from mismatched "in-batch negative" samples. This web-scale initialization structures the broader semantic boundaries of the vector manifold.

2. **Supervised Instruction Fine-Tuning:** To refine granular retrieval capabilities, the model undergoes secondary fine-tuning on high-quality, human-curated datasets (such as MIRACL and MS MARCO). Crucially, this stage introduces task-specific instruction prefixes (e.g., prepending "query: " or "passage: " to the input string). This instruction-aligned formatting teaches the model to reconcile structural asymmetries, ensuring that short, intent-driven query vectors map accurately to dense, informational document passages.

On standardized performance benchmarks such as MIRACL (Multilingual Information Retrieval Across a Continuum of Languages) and MTEB (Massive Text Embedding Benchmark), the model achieves top-tier results. The MTEB framework serves as the comprehensive global standard for text embedding evaluation, measuring performance across dozens of distinct datasets spanning seven unique downstream tasks including semantic similarity, classification, and retrieval. MIRACL focuses specifically on cross-lingual and ad-hoc retrieval capacity across 18 distinct languages, providing a granular assessment of non-English vector alignment.

Within these evaluations, the model's retrieval quality is primarily quantified using Normalized Discounted Cumulative Gain at a truncation position of 10 ($nDCG@10$), achieving a top-tier score of 66.5 on MIRACL. The $nDCG@10$ metric evaluates ranking quality by assessing the relevance of the top 10 retrieved documents while applying a logarithmic discount factor based on a document's specific position in the results list. This mathematical structure ensures that systems are heavily penalized for burying highly relevant documents near the bottom of the top-10 window, prioritizing immediate precision. However, these benchmarks evaluate retrieval in general-knowledge domains, such as Wikipedia hierarchies or

open-source web crawls. There remains a significant documentation gap regarding how these high-dimensional internal representations hold up when compressed into Int8 spaces for technical, legal, and administrative Dutch used in procurement.

While general performance remains high, the internal representations of compressed models are subject to what Proskurina et al. (2023) describe as the "other side of compression." Their research indicates that transformer-based encoders subjected to significant compression (such as pruning or aggressive quantization) can exhibit an amplification of internal biases. Even when the overall performance loss (e.g., nDCG@10) appears insignificant, the model may begin to over-rely on specific salient tokens or sensitive attributes at the expense of broader context.

This leads to a "calibration gap," where the model's predicted relevance score no longer accurately reflects its true probability of correctness. Proskurina et al. (2024) demonstrate that while 8-bit quantization is more stable than 4-bit, it still introduces fluctuations in model calibration across different scales. This is particularly relevant for the `e5-multilingual-large` model used here; as a larger model, it may possess more "salient neurons" that contribute to its high performance, but the impact of quantization on these specific neurons can vary depending on the task and language.

2.4 Domain-Specific Challenges: Dutch Procurement

Retrieval-Augmented Generation (RAG) and information extraction in specialized fields like construction or government tenders are sensitive to "out-of-distribution" terminology. Recent research in construction delay disputes has demonstrated that model performance is highly dependent on the nature of the input data (Wei et al., 2025). While previous studies have focused on model selection for classification, this research investigates the *precision-efficiency trade-off* in retrieval. The goal is to determine if the compression benefits of Int8 quantization are worth the potential loss in retrieval accuracy when the system must extract high-precision data from Dutch tender documents.

The introduction of "quantization-induced bias" and "confidence degradation" poses a risk for Dutch tender analysis. Since procurement documents often contain dense, technical, and administrative jargon, the "out-of-distribution" nature of the data already places the model in a lower-confidence regime. If, as Proskurina et al. (2024) suggest, quantization primarily degrades performance on these low-confidence samples, the resulting retrieval system might systematically fail on the most complex and often most critical sections of a tender.

This research, therefore, extends beyond a simple accuracy check. By measuring the precision-efficiency trade-off, this study essentially tests whether the compression techniques necessary for scaling dense retrieval systems preserve the reliability and faithfulness of the retrieval process in a linguistically specialized and legally sensitive environment.

2.5 Research Gap and Objectives

The current literature provides a clear understanding of the mechanics of vector retrieval and the general theory of Int8 quantization. However, the intersection of these technologies within a niche, high-stakes domain like Dutch procurement remains unexplored.

While general-purpose benchmarks suggest that Int8 quantization is highly efficient, there is a critical lack of research on whether this compression disproportionately affects specialized technical jargon compared to general industry vocabulary. In high-precision

retrieval, the loss of granularity inherent in 8-bit “binning” may cause the model to lose track of the salient neurons which are the specific internal experts responsible for identifying niche legal codes or administrative Dutch acronyms.

This research addresses this gap by moving beyond aggregate metrics toward a fine-grained, micro-level analysis, focusing on how vector precision constraints alter ranking stability and query behavior across different linguistic structures. Rather than evaluating quantization purely as a source of degradation, this study systematically analyzes the interplay between mathematical discretization noise, semantic density, and administrative Dutch vocabulary. This cross-lingual comparison aims to establish how 8-bit compression alters localized vector spaces, specifically isolating whether observed performance shifts are driven by language-specific syntax or the underlying technical complexity inherent to the procurement domain.

Chapter 3

Methodology

This section details the technical implementation and experimental setup used to evaluate the impact of Int8 quantization on semantic retrieval. As illustrated in the overarching architectural workflow in Figure 3.1, the methodology is structured as a bifurcated, parallel pipeline designed to isolate the effects of low-precision data representations across different linguistic contexts. The workflow systematically transitions from initial data preparation and cross-lingual translation to embedding generation, scalar quantization, and downstream rank evaluation.

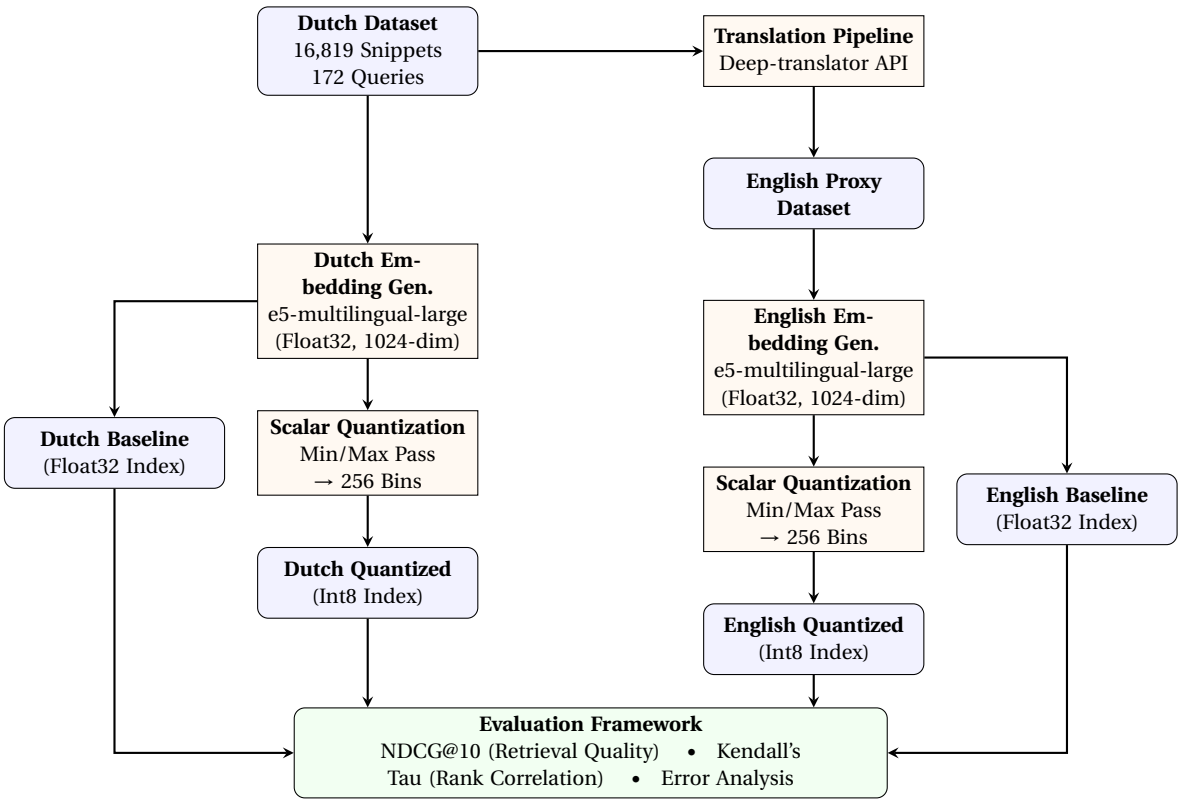


Figure 3.1: Methodological workflow showcasing parallel Dutch and English pipelines mapping from full-precision baselines to Int8 scalar quantized spaces.

3.1 System Blueprint Walkthrough

To thoroughly interpret the experimental mechanics, Figure 3.1 breaks down the operations into three core phases:

1. **Data Ingestion and Alignment:** The process initiates at the top row. The Dutch text snippets and queries are branched; one path feeds directly into the native processing pipeline, while the parallel path uses a machine translation gateway to construct an identical, structured English proxy corpus.

2. **Embedding and Discretization Dual-Tracks:** Both text corpora are mapped independently into identical dense vector spaces utilizing the `e5-multilingual-large` encoder. From this state, the full-precision `float32` vectors are cloned to serve as control parameters (the baselines), while the main operational path routes through a uniform min/max scalar quantization algorithm to yield experimental `int8` spaces.

3. **Downstream Metrics Aggregation:** Finally, all four decoupled topological spaces: the Dutch baseline, the Dutch quantized index, the English baseline, and the English quantized index, converge uniformly onto a unified evaluation engine. This component contrasts retrieval metrics and measures correlation degradation across both linguistic variants.

3.2 Corpus Architecture and Structural Document Profiling

The text repository used for this research was curated to evaluate the precision-efficiency trade-offs of embedding compression within a highly specialized, out-of-distribution domain. The foundational dataset consists of five large-scale, public sector procurement tenders published in the Netherlands. Rather than presenting simple, continuous text files, these five overarching tenders encompass a complex network of 169 unique, heterogeneous documents.

To visually trace how these raw administrative materials are programmatically transformed for our downstream embedding pipeline, Figure 3.2 outlines the complete structural workflow of our data ingestion.

As shown in the top row of Figure 3.2, these 169 source files are not evenly distributed among the 5 procurement projects. Instead, they introduce a distinct data skew. As formalized in Table 3.1, a single sprawling public project (Tender 1) accounts for nearly half of the entire document footprint ($n = 80$), whereas the remaining tender folders represent more compact administrative packages. This structural disparity is highly representative of real-world procurement operations, where specific projects necessitate massive libraries of technical addenda, and others remain constrained to core legal agreements.

Moving horizontally to the secondary tier of Figure 3.2, the middle layer illustrates that despite this numerical variance, the underlying documents map across five functional typologies. While manual document classification was not performed as a separate step for this study, the corpus natively represents these core administrative, legal, and operational segments as derived directly from the automated document indexing labels processed within the proprietary Brainial B.V. infrastructure:

- **Gunningsleidraden:** Official procurement guidelines establishing the explicit rules, evaluation criteria, and award scoring systems governing the project.

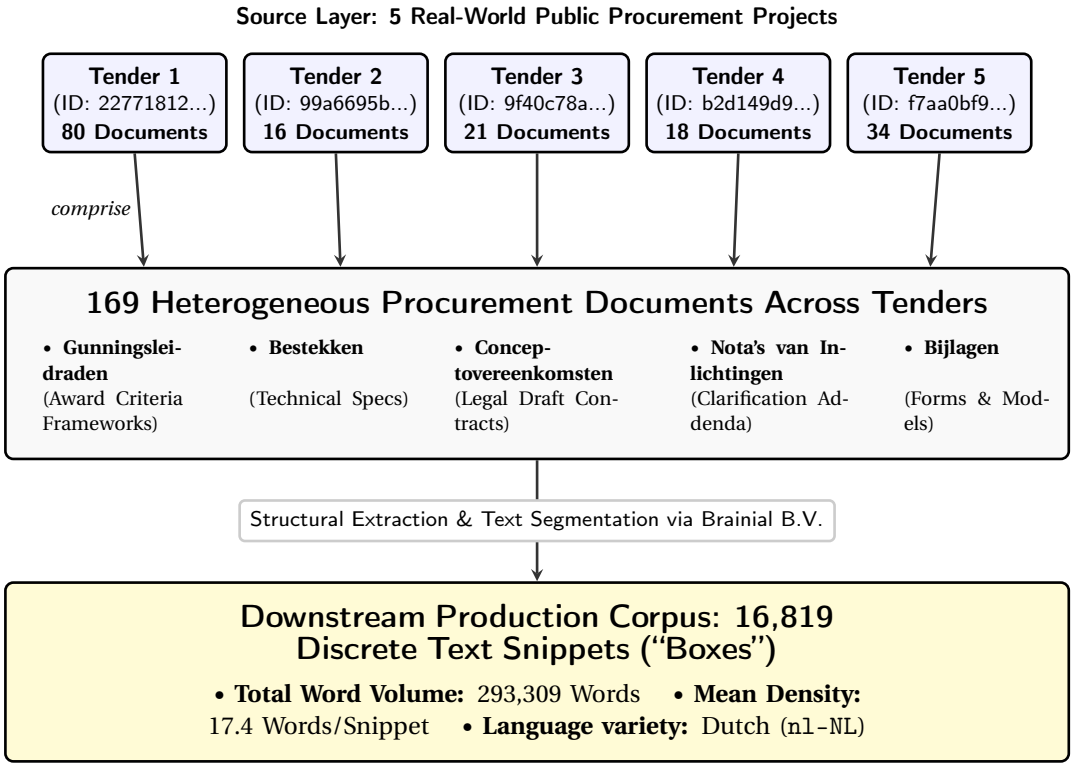


Figure 3.2: Structural workflow of data ingestion demonstrating the uneven macro distribution of documents across the five public tenders and the subsequent programmatic micro-segmentation into discrete vector tracking snippets.

- **Bestekken:** Comprehensive technical specification portfolios detailing rigorous material standards, design norms, and physical construction requirements.
- **Conceptovereenkomsten:** Formal draft purchasing contracts documenting structural legal liabilities, penalty constraints, and commercial validation bounds.
- **Nota's van Inlichtingen:** Highly critical clarification addenda capturing statutory corporate questions and answers published throughout the tender lifecycle.
- **Bijlagen:** Supplementary appendices containing mandatory declaration forms, regional geographical scopes, and explicit pricing models.

The foundational transition connecting Layer 2 to Layer 3 in Figure 3.2 represents the primary programmatic data extraction pipeline. Because dense semantic models natively require compact text limits rather than full multi-page layouts, the unstructured document fields were ingested and parsed into granular segments using the advanced layout-parsing software from Brainial B.V. This processing block successfully isolated a total of 16,819 discrete text snippets, referred to within our retrieval framework as "boxes". To safeguard commercial intellectual property, the underlying proprietary algorithms governing Brainial's layout-parsing engine are omitted from this text.

Ultimately, this structural workflow yields the final production corpus outlined at the base of our funnel diagram. Cumulatively, the parsed text blocks span a total of 293,309 words, sustaining a tight density distribution with a mean length of 17.4 words per individual chunk. This final dataset functions as a highly representative linguistic corpus for real-

Table 3.1: Distribution of Unique Documents Across the Five Procurement Tenders

Tender Reference Identifier	**Number of Documents**
Tender 1 (`22771812-c715...`)	80
Tender 2 (`99a6695b-33b0...`)	16
Tender 3 (`9f40c78a-76c9...`)	21
Tender 4 (`b2d149d9-b77e...`)	18
Tender 5 (`f7aa0bf9-ddf8...`)	34
Total Corpus Volume	**169**

world procurement automation, heavily saturated with domain-specific compound terms, alphanumeric specifications, and non-standard layout formatting anomalies that uniquely challenge compressed vector models.

3.3 Quantization Strategy and Density Profile

As introduced in the background review of vector quantization (Section 2.2), uniform scalar quantization can be conceptualized as slicing a smooth, continuous spectrum of meaning into 256 discrete categories. However, the linguistic impact of this process depends entirely on how the model naturally distributes words and sentences across this spectrum.

While the `e5-multilingual-large` model is theoretically capable of spreading data out across a wide range from -1.0 to 1.0, our specialized Dutch tender data behaves quite differently. In practice, the embedding space is highly crowded; nearly all text snippets are compressed into a microscopic window ($\sigma = 0.03$), clustered tightly together. This extreme concentration creates a significant roadblock when trying to convert these values into a simplified grid.

Because our quantization approach spaces its 256 bins evenly across the entire theoretical range, the vast majority of these bins are left completely empty, waiting for extreme statistical outliers that rarely appear. Meanwhile, the dense cluster containing almost all of our actual tender data is forced to share only a tiny fraction of the available bins. The resulting bin width (roughly 0.0016 units) establishes a strict limit on the model's linguistic resolution.

Consequently, any subtle semantic nuance smaller than this width is completely erased when forcing a 32-bit floating-point number (`float32`) into an 8-bit integer (`int8`). In procurement documents, this lost nuance is precisely where the critical meaning hides. For example, the subtle lexical differences that distinguish one specific legal requirement from another can be lost. Flattening these values causes a sort of "semantic blurring," which directly explains why the system mixes up the order of relevant search results later on.

The practical boundaries and mechanics of how this distribution handles Dutch procurement text are detailed below:

- **Equal-Width Discretization Mechanics:** This approach treats the spectrum of meaning like a standard ruler, where every single mark is separated by an identical distance. While advanced computational literature often uses uneven, flexible boundaries to protect rare words and sudden statistical spikes (Dettmers et al., 2022), a rigid, uniform grid was chosen for this research. This decision directly accommodates industrial database engines like Elasticsearch, which process simple grids at lightning-fast speeds, even if it means sacrificing the model's ability to see fine linguistic detail.

- **Empirical Density Clustering and the Paint-by-Numbers Constraint:** Although the system prepares for values spanning from a minimum of -1.0 to a maximum of 1.0, our tender data actually huddles almost exclusively within a tiny window between -0.13 and 0.27. To understand the impact of this distribution, imagine a "paint-by-numbers" kit. The kit comes with 256 numbered paint pots meant to cover every color in the universe. However, because tender documents use highly specific, repetitive procurement language, our entire "masterpiece" consists only of ultra-subtle variations of corporate blue. Instead of spreading those 256 paint numbers across our blues to capture every fine detail, the rigid system wastes most of its numbers on vibrant reds, yellows, and greens that our text never uses. We are left trying to paint a highly detailed sky using only five numbered pots, forcing distinct shades to share the exact same number.
- **Linear Meaning Mapping:** Every highly precise coordinate in the original model is forced to snap to the closest 8-bit integer category ($i \in \{-128, \ldots, 127\}$). In our artistic analogy, this is the exact moment the brush is forced to stop blending custom colors and must strictly follow the boundary lines printed on the paint-by-numbers canvas.
- **Precision Loss and Linguistic Collapse:** Snapping values to fixed points collapses billions of possible variations into a mere 256 bins. This loss of grain is the main culprit behind the retrieval degradation documented in this study. When multiple distinct technical terms are forced to use the exact same paint number, the computer can no longer tell them apart, directly explaining the variance in retrieval quality (`NDCG@10`) and ranking consistency (`Kendall's Tau`) evaluated in Section 3.5.

3.4 Memory and Computational Impact

By transitioning from continuous values to the rigid, numbered grid detailed in the quantization strategy of Section 3.3, the system achieves massive computational efficiency. This "paint-by-numbers" simplification reduces the memory footprint of the 1024-dimensional vectors from 4,096 bytes to 1,024 bytes, representing a 75% reduction in raw storage requirements.

This drastic physical transformation is critical for the scalability of industrial search infrastructure, specifically within the Elasticsearch `dense_vector` field. By stripping away the infinitely complex decimal spaces of the original model, it allows the Hierarchical Navigable Small World (HNSW) algorithm (see 2.1) to perform its vast nearest-neighbor distance calculations using basic integer arithmetic. For a database engine, comparing simple whole numbers is significantly faster and less hardware-intensive than executing billions of complex floating-point calculations (`float32`).

This stark efficiency gain addresses the practical constraints of balancing high retrieval accuracy with corporate computational costs. By compressing the bit-precision from 32-bit to 8-bit, the system significantly lowers both the active RAM usage and the computational energy required to execute vector similarity searches, contributing directly to more sustainable and commercially viable AI practices.

3.5 Evaluation Framework

With both the baseline and quantized indices established for the primary data, the research proceeds to the performance measurement phase. As highlighted at the base terminal of

Figure 3.1, the downstream evaluation does not rely on a single, generic accuracy score. Instead, it utilizes a dual-metric framework designed to isolate two distinct dimensions of retrieval degradation: absolute ranking quality and hidden internal space shifting.

3.5.1 Normalized Discounted Cumulative Gain (NDCG@10)

Conceptual Intuition

From a linguistic and information retrieval perspective, `NDCG@10` measures the immediate, visible utility of a search engine to a human user. In production environments like Elasticsearch, a user rarely looks past the first page of results. Therefore, this metric evaluates the quality of the top 10 retrieved text snippets based on two common-sense assumptions:

1. **Cumulative Gain:** Highly relevant documents are more useful than partially relevant documents, which are far more useful than completely irrelevant ones.

2. **Discounting:** A highly relevant document is vastly more valuable if it appears at the very top of the results page (Position 1) than if it is buried near the bottom (Position 10).

While global benchmarks such as MTEB and MIRACL utilize `NDCG@10` to rank models on general-knowledge data like Wikipedia crawls (where the `E5-Multilingual-Large` model achieves a top-tier score of 66.5), this study deploys it to measure how heavily a lossy 8-bit grid blurs or penalizes the position of domain-specific Dutch procurement data. If quantization causes a vital legal requirement to drop from Rank 1 to Rank 9, `NDCG@10` will reflect this degradation.

How It Is Calculated

The calculation of NDCG@10 can be understood as a three-step logical pipeline:

- **Step 1 (Gathering Gain):** The system looks at the top 10 results and counts how many are true matches based on our expert human "Gold Standard" labels. In our setup, a verified match gets a score of 1, and an irrelevant snippet gets a 0. This relevance score for a document at rank i is denoted as rel_i.

- **Step 2 (Applying the Position Penalty):** The system divides each document's score by a logarithmic penalty based on its rank. Position 1 receives virtually no penalty, while Position 10 receives a much larger penalty. This ensures that burying a critical document at the bottom heavily damages the score. Specifically, the logarithmic discount factor exponentially diminishes the document's weight as its rank increases. For instance, a perfect match found at Rank 10 contributes less than a third of its potential value compared to if it were placed at Rank 1. Summing these penalized scores yields the *Discounted Cumulative Gain* (DCG@10):

$$\text{DCG@10} = \sum_{i=1}^{10} \frac{rel_i}{\log_2(i+1)}$$

- **Step 3 (Normalization):** To make sure scores are fair across different queries—since some search queries naturally have more matching documents in the archive than others—the actual search engine score is divided by an "Ideal" score. This *Ideal DCG*

(IDCG@10) is what a perfect search engine would get if it sorted the available documents flawlessly from best to worst:

$$\text{IDCG@10} = \sum_{i=1}^{10} \frac{rel_i^{\text{ideal}}}{\log_2(i+1)}$$

By dividing the actual performance by the ideal performance, we get the final formula:

$$\text{NDCG@10} = \frac{\text{DCG@10}}{\text{IDCG@10}}$$

The final score scales cleanly between 0.0 and 1.0, where 1.0 represents absolute perfection.

3.5.2 Kendall's Tau (τ) Rank Correlation

Conceptual Intuition

While `NDCG@10` strictly isolates what happens at the absolute surface level of the top 10 results, it is a blunt tool for studying the internal behavior of a vector space. If a cluster of fifty documents violently shifts positions, but they all remain outside the top 10 window, `NDCG@10` registers a score change of exactly zero. To catch this hidden, internal shuffling caused by Int8 compression, we implement Kendall's Tau (τ).

Conceptually, Kendall's Tau acts as a comprehensive rank correlation metric. It takes the entire list of results returned by the full-precision (`float32`) model and compares it side-by-side with the list returned by the compressed (`int8`) model. It looks at every single possible pair of documents and asks a simple question: *"Did the computer maintain the relative order of these two items after the numbers were simplified, or did they swap places?"*

If the full-precision model originally decided Document A was slightly more relevant than Document B, Kendall's Tau verifies whether that specific relationship survived the quantization process. It tells us how structurally stable the model's understanding of meaning remains under precision constraints.

How It Is Calculated

Kendall's Tau calculates its final score by analyzing the directional agreement of document pairs, treating the entire results list like a series of head-to-head matchups:

- **Concordant Pairs (Agreements, P):** If Document A is ranked higher than Document B in both the high-resolution and the compressed list, the pair is marked as an "agreement."
- **Discordant Pairs (Disagreements, Q):** If Document A was higher than Document B in the high-resolution list, but they flip-flop and Document B becomes higher in the compressed list, it is marked as a "disagreement."
- **Handling Ties (T_X and T_Y):** Because our 256-pot "paint-by-numbers" grid forces thousands of precise decimals to collapse into identical integer buckets, many documents end up with identical similarity scores. To address this, we utilize the standard `scipy.stats.kendalltau` implementation, which defaults to the *Kendall's Tau-b* variant. This specific version mathematically adjusts for tied brackets in the full-precision list (T_X) and the compressed list (T_Y) so they do not artificially inflate the consistency score.

The final score is calculated using the following formula:

$$\tau = \frac{P - Q}{\sqrt{(P + Q + T_X)(P + Q + T_Y)}}$$

This scales the metric between -1.0 (a perfect, complete reversal of the list) and $+1.0$ (an uncompromised, perfectly preserved rank structure), where 0.0 indicates entirely random, chaotic ordering.

Justification for Selection Over Alternative Metrics

While traditional metrics like Pearson and Spearman rely on raw numerical values or squared rank distances, they fail to handle the non-linear shifts and massive volume of score ties caused by Int8 quantization. Kendall's Tau is uniquely suited for this scenario because its pairwise, "head-to-head" comparison mechanic tracks localized rank flips and mathematically corrects for tied integer buckets without skewing the structural stability score. An overview is shown in Table 3.2

Metric	What It Measures	Why It Fails under Quantization
Pearson	Strict linear numerical value relationships.	Fails completely because search results are ordered ranks (1st, 2nd, 3rd), not continuous numbers. Shifts do not scale linearly.
Spearman	Distance gaps between ranks using squared math.	Explodes and becomes erratic due to isolated extreme jumps, and cannot handle mass score ties gracefully.
Kendall's Tau	Relative, pairwise "head-to-head" order agreement.	Safely penalizes and tracks tied bins caused by our 256-box limitation without letting outliers hide systemic errors.

Table 3.2: Comparative analysis of correlation metrics under Int8 quantization constraints.

3.6 Cross-Lingual Translation Experiment Setup

To evaluate whether the impacts of Int8 quantization are language-dependent or tied to the unique linguistic structures of the Dutch language, a cross-lingual experimental setup was introduced. By translating the dataset, the exact same processing and quantization pipeline applied to the Dutch data could be reused sequentially for an equivalent English corpus. This allows for a direct, isolated comparison between the quantization behavior on the Dutch dataset and its English proxy.

3.6.1 Dataset Translation Pipeline

Because a original English validation dataset with domain-specific labels was unavailable, an English proxy corpus was constructed using automated machine translation. The translation workflow was executed in Python using the `deep_translator` library, leveraging the

Google Translate API. The pipeline processed the data in two discrete phases, represented by the "Translation Pipeline" system block in Figure 3.1:

- **Query Translation:** All 172 Dutch "Points of Attention" (POAs) used as research queries were translated into English.
- **Corpus Translation:** The 16,819 text snippets ("boxes") spanning the 169 unique tender documents were translated entirely into English, preserving their original structural mapping and unique identifiers.

To verify the fidelity of the automated translation, a manual spot-check was performed on a random selection of translated pairs. This qualitative validation was conducted by the author of this thesis (a native Dutch speaker pursuing an English-taught Master's degree) possessing professional proficiency in both languages. The check confirmed that specialized technical procurement terms and contextual nuances were preserved accurately, with no observable loss of semantic meaning.

Given the scale of the corpus, a full manual review of all 16,819 snippets was impractical. Therefore, the existing "Gold Standard" ground-truth mapping established by the senior tender manager was mapped directly to these translated pairs, operating under the grounded assumption, and supported by the spot-checks, that semantic relevance remains invariant under accurate translation.

3.6.2 Experimental Execution

Following data translation, the English corpus was subjected to the exact same technical pipeline detailed in the preceding sections:

1. **English Embedding Baseline:** The translated text snippets and queries were encoded using the `e5-multilingual-large` model at native `float32` precision, generating a parallel 1024-dimensional English baseline vector space.
2. **English Scalar Quantization:** Dynamic range estimation was performed on the English embeddings to establish empirical minima and maxima, followed by linear spacing discretization into 256 integer bins (`int8`).
3. **Downstream Evaluation:** The resulting full-precision and quantized English embedding indices were passed to the evaluation framework detailed in Section 3.5 to compute retrieval metrics, allowing for a side-by-side comparative analysis of semantic compression across both languages.

This cross-lingual setup isolates the effect of quantization from the underlying language, highlighting whether the observed precision losses are systemic to the `e5-multilingual-large` architecture or exacerbated by the specific nuances of technical Dutch procurement terminology.

3.7 Synthetic Data Experiment for Special Characters

To understand exactly why the system sometimes struggles with specific search queries, a controlled "stress test" was designed. While the system performs well on average, these broad scores can hide the fact that certain symbols or formatting can cause the retrieval process to fail on specific documents. Initial observations suggested that the model is highly

sensitive to non-textual elements, such as emojis or currency symbols, when the data is compressed.

This experiment was designed to determine if these specific symbols actually force the model to make errors, regardless of the surrounding text's meaning. By keeping the sentence meaning exactly the same but adding or removing symbols, it becomes possible to pinpoint whether these items break the model's underlying math. The results of this test, and how these symbols change the model's internal calculations, are detailed in Chapter 4.4.

The test dataset consists of 50 distinct sentences organized into 25 evaluation pairs, which were manually compiled and engineered by the author based on real-world Dutch procurement texts. This sample size of 50 sentences was chosen as a pragmatic number to ensure a diverse cross-section of token types while operating within the time and resource constraints of this thesis project. The primary target baseline text (P00) was established using a common procurement constraint: *"The maximum contract value for this project is capped at 500000..."*. For each of the 25 configurations (the raw pairs of which are documented in Appendix .1), a control string consisting purely of alphanumeric characters was paired against an identical semantic twin injected with a localized, non-textual structural token. For example, configuration `P01` contrasts the clean baseline with a currency-injected twin:

> **Control (Alphanumeric):** The maximum contract value for this project is capped at 500000."
> **Treatment (Injected Twin):** The maximum contract value for this project is capped at €500000."

Across these 25 pairs, the token types included:

- **Warning Identifiers:** Standard warning icons (applied to pairs P00, P02, P05, P08, P11, P13, P16, P19, P23, and P24) or stacked dual warning icons (P20).
- **Operational Markers:** Action blocks indicating rigid parameters, such as a red circle (P03, P06, P14, P21), a deletion cross (P07, P15), or a green validation checkmark (P04, P12, P17, P22).
- **Financial Notation Keys:** Currency indicators, spanning isolated cash symbols (P09) or explicit euro symbol prefixes inserted directly before numeric criteria (*€500000* in P01, *€750000* in P10, and *€2000000* in P18).

The entire synthetic corpus was evaluated against a clean semantic query using the original 32-bit floating-point precision (*Float32*) and a parallel 8-bit quantization (*Int8*) configuration. The full tabular log records for this setup are compiled in Appendix .3. The overall processing mechanics of this evaluation methodology are mapped out step-by-step in Figure 3.3.

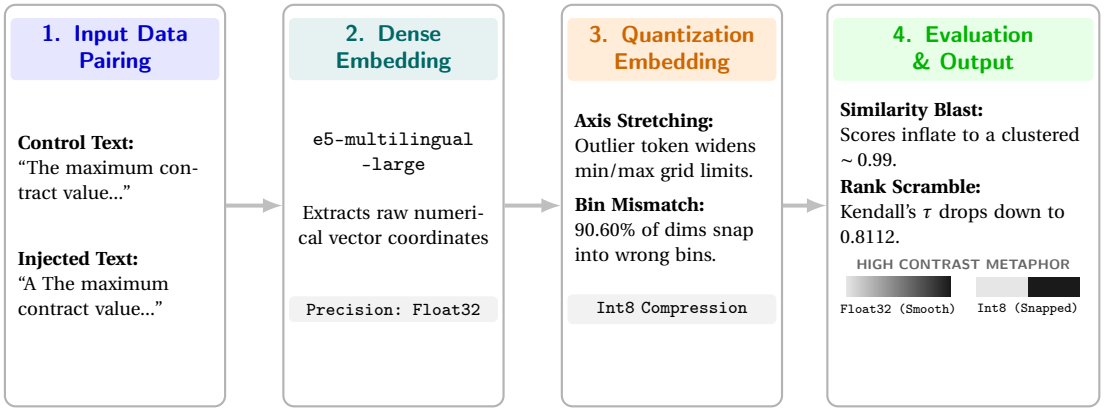


Figure 3.3: Methodological processing pipeline for the synthetic emoji stress test. It illustrates the progression from textual injection to dense vector generation, quantization of the embedding, and final rank degradation.

Chapter 4

Results

4.1 Dutch Retrieval Performance Evaluation

The evaluation was conducted on a dataset comprising 172 Dutch procurement queries, each benchmarked against a candidate pool of 100 randomly sampled document snippets (boxes). To ensure a rigorous evaluation of ranking capabilities, all gold-standard documents were guaranteed to be present within the candidate set. Two retrieval configurations were compared: a baseline utilizing full-precision float embeddings and an `Int8`-quantized variant, both derived from the same `multilingual-e5-large` ONNX model.

4.1.1 Aggregate Retrieval Quality

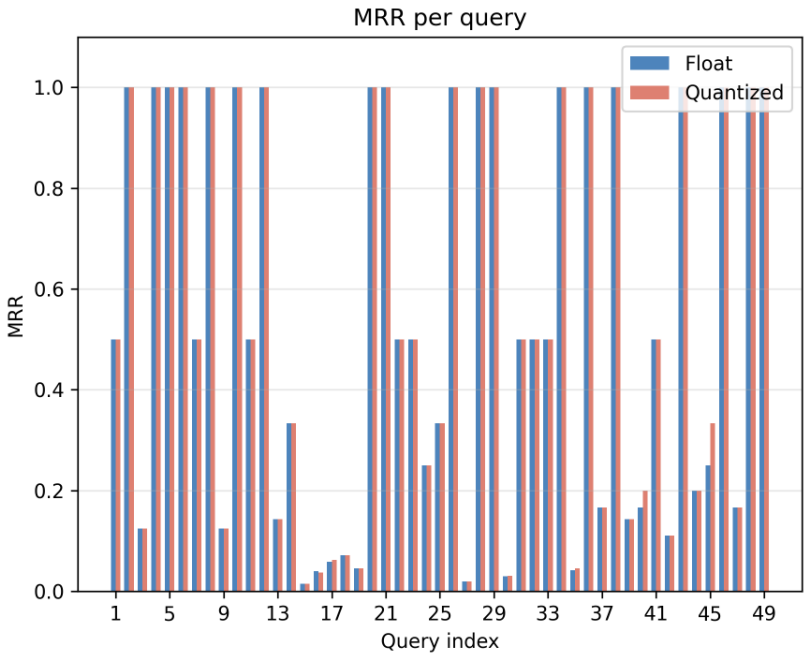
As illustrated in the summary block provided in Figure 4.1, the two configurations achieve nearly identical performance across standard retrieval metrics. The mean NDCG@10 scores are 0.5969 for the float baseline and 0.5987 for the quantized variant, showing a negligible delta of less than 0.002. Mean Reciprocal Rank (MRR) follows a similar parallel trajectory at 0.6410 and 0.6395, respectively. At the macro level, these evaluation metrics indicate that `Int8` quantization introduces no statistically meaningful degradation in retrieval quality across this specialized domain.

```
Metric                 Float  Quantized
-------------------------------------
Mean NDCG@10          0.5969     0.5987
Mean MRR              0.6410     0.6395
Mean Kendall τ        0.9882 (pairwise)
```

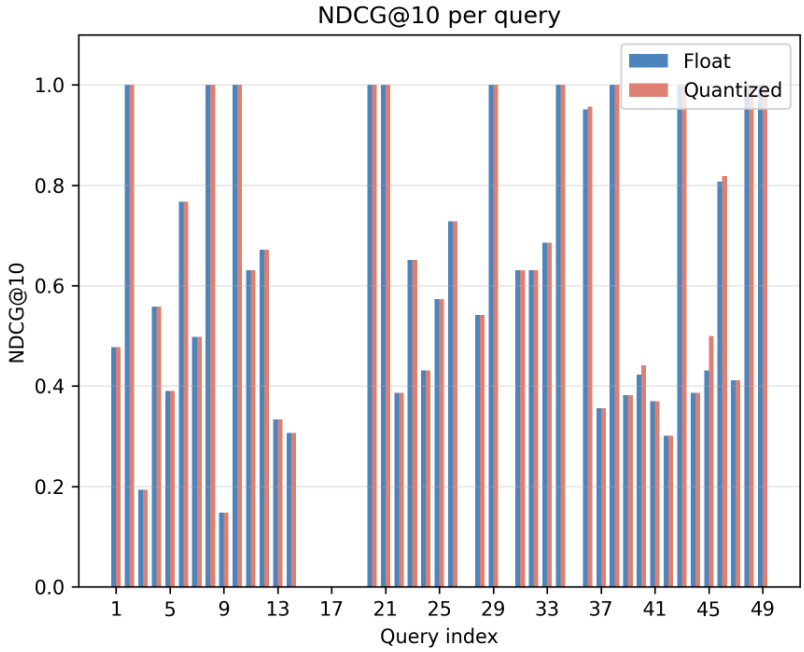
Figure 4.1: Macro Performance Metrics: Comparison ledger of aggregate retrieval quality between full-precision baseline and Int8 scalar quantized configurations for the Dutch dataset.

4.1.2 Per-Query Variation and Stability

While aggregate metrics suggest structural stability, the expanded query-by-query performance breakdowns in Figure 4.2 reveal that macro averages mask localized variations at



(a) Per-Query Reciprocal Rank (MRR)



(b) Per-Query Retrieval Quality (NDCG@10)

Figure 4.2: Micro-Level Query Distributions: Expanded individual query (first 50 queries) index distributions for MRR and NDCG@10, highlighting the highly resilient score overlap between the baseline float and compressed paths.

the individual query level. Of the 172 queries evaluated, quantized embeddings produced a strictly lower NDCG@10 score on 4 queries and a strictly higher score on 13 queries, while the remaining 155 queries yielded completely identical results.

For optimal visual readability within the plotting layout, the individual query breakdowns for MRR and NDCG@10 depicted in Figure 4.2, as well as the corresponding rank correlation tracking line illustrated in Figure 4.3, display a representative subset containing the first 50 query indices. Across the entire 172-query corpus, the pairwise rank correlation remains consistently high, tightly bounded between 0.980 and 0.995 with a mean of 0.988. This narrow range confirms that the relative ordering of documents is highly stable under compressed constraints, with no isolated query index experiencing any significant reordering of its candidate document pool.

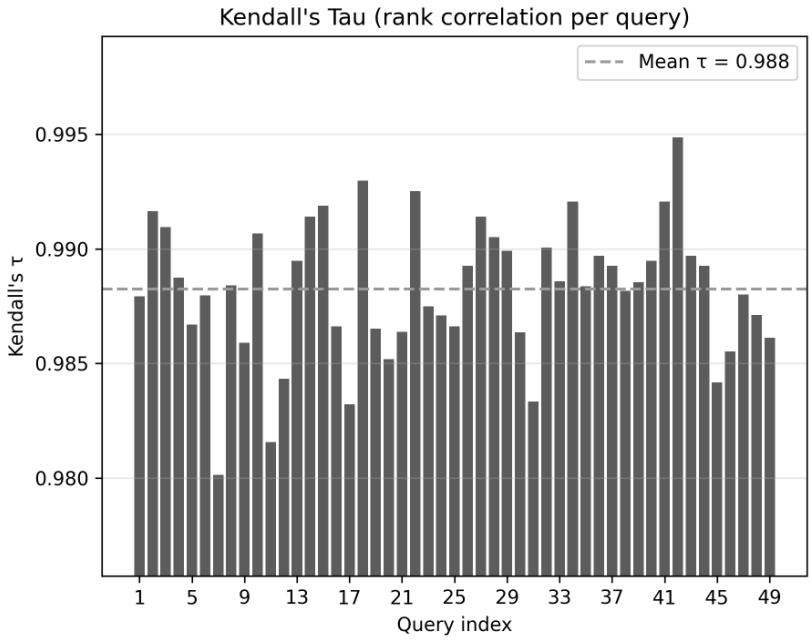
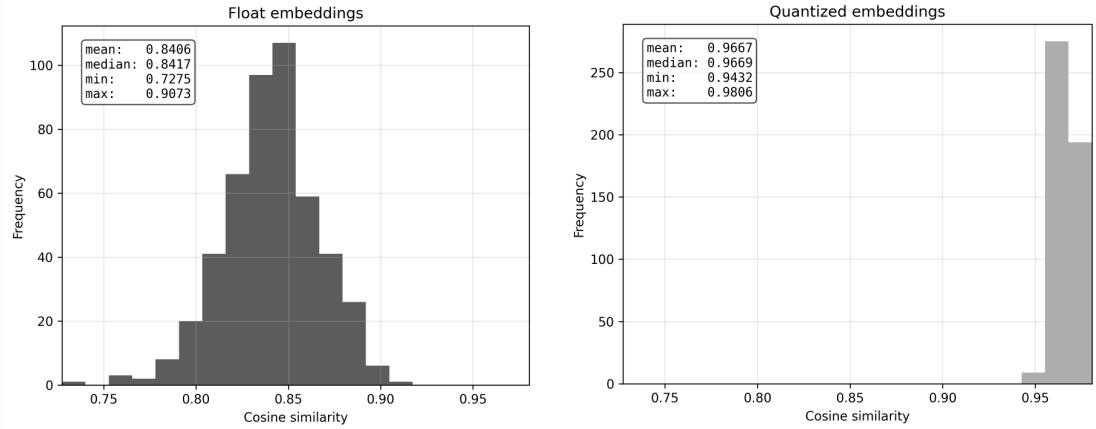


Figure 4.3: Pairwise Rank Stability Tracking: Kendall's τ correlation scores per individual search query (for the first 50 queries), demonstrating that the structural ordering of documents remains virtually uncompromised after matrix compression.

4.1.3 Score Distribution and Quantization Artifacts

A distinct geometric artifact of the compression scheme is visible when comparing the structural changes between the baseline distribution and the quantized space shown in Figure 4.4. The mean cosine similarity of matching gold documents aggressively shifts rightward from 0.841 to 0.967, representing an absolute inflation of approximately 0.13 points.

This rightward density shift contracts the dynamic range of the similarity scores into a concentrated distribution window. Stripping away the fine-grained dimensional gradients that originally distinguished individual text chunks causes the relative distance between different document vectors to narrow significantly. When these quantized vector representations are re-normalized to calculate cosine similarity scores, the absolute similarity magnitudes cluster tightly near a value of 1.0, rendering absolute similarity thresholds incomparable across the two precision spaces.



(a) Dutch Float Cosine Similarity (b) Dutch Quantized Cosine Similarity

Figure 4.4: Geometric Similarity Scaling and Inflation: Frequency distributions comparing full-precision floating-point similarities against the Int8 compressed space, highlighting the radical rightward density shift toward 1.0.

Table 4.1: Dutch queries with lower NDCG@10 under quantized embeddings, including full literal and contextual translations

Query	**Topic (with English Translation)**	**Float**	**Quant**	Δ
Q107	Maximale opdrachtwaarde/plafondbudget *(Maximum contract value / ceiling budget)*	0.5585	0.3973	−0.1612
Q53	Harde uitsluitingseis *(Hard exclusion criteria / mandatory requirement)*	0.9197	0.8772	−0.0425
Q16	Max. aantal percelen aan inschrijver *(Maximum number of lots per bidder)*	0.8285	0.8216	−0.0070
Q104	Aantal vereiste referentieprojecten *(Number of required reference projects)*	0.7157	0.7087	−0.0070

Table 4.2: Dutch queries with higher NDCG@10 under quantized embeddings, including English translations

Query	**Topic (with English Translation)**	**Float**	**Quant**	Δ
Q31	Boeteregeling / hoogte boete *(Penalty scheme / fine amount)*	0.0000	0.2891	+0.2891
Q132	Max. pagina's/slides Plan van Aanpak *(Maximum pages/slides Action Plan)*	0.4307	0.5000	+0.0693
Q5	Gunningscriteria / subgunningscriteria *(Award criteria / sub-award criteria)*	0.6502	0.6968	+0.0465
Q36	Overmachtsbepalingen (force majeure) *(Force majeure provisions)*	0.4306	0.4575	+0.0269
Q17	Geografisch werkgebied *(Geographical working area / scope)*	0.2184	0.2372	+0.0188
Q152	Gestanddoeningtermijn (dagen) *(Validity period / period of tender in days)*	0.4228	0.4416	+0.0188
Q124	Geschatte vs. maximale opdrachtwaarde *(Estimated vs. maximum contract value)*	0.7246	0.7366	+0.0120
Q140	Indexeringsregeling / prijsverhoging *(Indexation scheme / price increase terms)*	0.8072	0.8180	+0.0107

Despite this intense numerical crowding, the absolute metric changes reported in Section 4.1.1 remain minimal. This macro-level stability indicates that while uniform scalar quantization compresses values into a narrower numeric spectrum, it largely preserves the monotonic relative ordering required for nearest-neighbor rank evaluation. The underlying geometric and structural implications of this coordinate crowding phenomenon on specialized language spaces are explored extensively in Chapter 5.

4.1.4 Failure Case Analysis and Qualitative Tracking

A subset of the queries proved challenging for both configurations. Out of the 172 queries evaluated, 22 (12.8%) yielded an NDCG@10 of zero, indicating that neither method ranked a gold document within the top 10. Furthermore, 57 queries (33.1%) produced an MRR below 0.5, meaning the first relevant document appeared outside the top two positions.

On average, gold documents were ranked in the top 10 in 71.3% of cases (341 out of 478 individual gold documents), with a mean rank of 11.5 and a median rank of 5.0. However, the maximum observed rank reached 98 out of 100, demonstrating that retrieval failures, while infrequent, can be quite severe. Rather than relying on solely intuition, these empirical outliers served as the direct catalyst for the stress test designed in Section 3.7. These retrieval blind spots empirically highlighted a systematic vulnerabilities confirming the need to isolate how specific structural noise disrupts the model's math.

To understand the micro-level mechanics operating beneath these aggregate performance statistics, a fine-grained tracking log analysis was conducted on the Dutch pipeline. This section highlights two routine administrative target queries ($Q1$ and $Q2$) that demonstrate the precise rank transitions and scoring behavior that occur when shifting from continuous spaces to discrete 8-bit integers:

- **Explicit Temporal Mapping (Q1 – Timeline Criteria):** Query 1 (*"Wat is de totale duur van het contract?"* / "What is the total duration of the contract?") achieved identical retrieval metrics (NDCG@10 = 0.6309; MRR = 0.5000) across both representation formats. In both cases, the model successfully found the exact gold text segment containing the timeline details at rank 2. However, while the ranking remained stable, the similarity scores showed the typical inflation pattern, jumping from $s_{\text{float}} = 0.8320$ to $s_{\text{int8}} = 0.9650$. This example validates the idea that absolute score shifts do not necessarily ruin retrieval, because the relative ordering of the documents is preserved. Even though the uniform scalar quantization compresses the embeddings into fewer discrete bins, it keeps the boundaries between candidates intact. Because straightforward chronological parameters do not create vector outliers, the margin between the true target and its competitors remains stable. This supports the theory that 8-bit quantization retains enough resolution to map clear numeric data safely.

- **Contextual Overlap and Semantic Drift (Q2 – Termination Windows):** Conversely, Query 2 (*"Wat is de opzegtermijn van het contract?"* / "What is the notice period of the contract?") illustrates how dense retrieval models struggle with overlapping legal text. Both the float and quantized configurations put incorrect clauses at rank 1 with high similarity scores ($s_{\text{float}} = 0.8192$; $s_{\text{int8}} = 0.9623$), while pushing the actual notice window further down the results list. These false positives, specifically rules for instant dismissal and 30-day billing cycles, are contextually related because they share a dense vocabulary of temporal constraints (such as "days," "periods," or "immediate") and contract termination concepts. This failure tracking logs a distinct baseline

alignment anomaly where semantic similarities are tightly clustered under both full-precision and low-precision constraints. The deep underlying morphological causes of this behavior, alongside tokenization bottlenecks and language-specific subword representation mechanics, are evaluated critically within Chapter 5.

4.1.5 Micro-Level Degradations and Performance Gains

While the aggregate metrics established in Section 4.1.1 indicate global stability, query-by-query examination reveals micro-level performance shifts when transitioning to low-precision integer spaces. This section breaks down individual Dutch queries where the `Int8` scheme induced localized performance regressions (detailed in Table 4.3) or, conversely, triggered visible retrieval improvements (detailed in Table 4.4).

Analysis of Localized Regressions

Table 4.3: Dutch queries with lower NDCG@10 under quantized embeddings, including full literal and contextual translations

Query	Topic (with English Translation)	Float	Quant	Δ
Q107	Maximale opdrachtwaarde/plafondbudget *(Maximum contract value / ceiling budget)*	0.5585	0.3973	−0.1612
Q53	Harde uitsluitingseis *(Hard exclusion criteria / mandatory requirement)*	0.9197	0.8772	−0.0425
Q16	Max. aantal percelen aan inschrijver *(Maximum number of lots per bidder)*	0.8285	0.8216	−0.0070
Q104	Aantal vereiste referentieprojecten *(Number of required reference projects)*	0.7157	0.7087	−0.0070

- **Q107 (Maximale opdrachtwaarde/plafondbudget [Maximum contract value / ceiling budget]):** This query suffered the most significant regression ($\Delta = -0.1612$). Error log analysis reveals that under the `Int8` scheme, the system mistakenly assigned the highest ranking to a generic form text box containing the boilerplate text: *"⚠ Indien u de vorige vraag met 'Ja' hebt beantwoord..."* (*"⚠ If you answered 'Yes' to the previous question..."*). This systematic error suggests a qualitative inference: the inclusion of non-alphanumeric structural tokens such as the warning emoji (⚠), introduces localized noise during vector quantization. This token disruption appears to mask the subtle adjacent numeric coordinates necessary to isolate explicit monetary limitations, causing the compressed model to prioritize boilerplate page elements over legal substance. This specific vulnerability and its exact underlying mathematical scaling mechanics are isolated and verified empirically via the controlled synthetic stress test in Section 3.7.

- **Q53 (Harde uitsluitingseis [Hard exclusion criteria / mandatory requirement]):** Performance decreased by −0.0425. While initial metrics indicate a performance delta between precision tiers, log file tracking reveals that this regression is an artifact of evaluation dataset annotation rather than a failure of integer quantization. In a procurement setting, mandatory knock-out criteria are frequently repeated across mul-

tiple sections using variable terminology, creating high semantic redundancy. Under the compressed `Int8` space, the model successfully isolated valid exclusion mechanisms but was penalized by strict metric boundaries. Specifically, the system retrieved an equally valid, active structural penalty clause regarding the prohibition of conditional bids: *"Het is niet toegestaan om een voorwaardelijke inschrijving te doen... zal de inschrijving worden uitgesloten van deelname"* (*"It is not permitted to submit a conditional bid... the bid will be excluded from participation"*). However, because the human annotator omitted this specific text block from the gold labels for this query, while restricting the gold standard to a narrow subset of alternative passages, such as the clause explicitly detailing pricing sheet modifications: *"Het is niet toegestaan om wijzigingen aan te brengen in Bijlage D en/of E... kan dit uitsluiting... betekenen"* (*"Het is not permitted to make changes to Appendix D and/or E... this may mean exclusion..."*), this high-precision retrieval was flagged as a false positive. The rank swap observed in the metrics does not demonstrate a degradation of semantic capacity caused by quantization, but rather exposes the incomplete coverage of the evaluation's gold standard when encountering redundant compliance declarations.

- **Q16 & Q104:** Both queries exhibited minor, negligible score reductions of -0.0070. Because these trace drops reflect a single document shifting down by merely one or two positions without impacting overall retrieval utility, isolating an exact causal layout factor or structural token dependency falls outside the analytical scope of this study. These fractional variances are transparently acknowledged as baseline mathematical noise. They represent the practical precision limit of compressing 32-bit floats down to 8-bit integers, verifying that while the overall ranking remains highly robust, trace fluctuations can occur independently of explicit text patterns.

Analysis of Localized Regularization Gains

Counterintuitively, a larger number of queries exhibited improved retrieval scores under quantized embeddings, as summarized in Table 4.4. This occurs when quantization acts as a semantic regularizer, smoothing away minor embedding noise and preventing overfitting to hyper-specific phrasing. To unpack this phenomenon, the analysis below elaborates on the three query examples that yielded the largest metric improvements across the dataset.

Table 4.4: Dutch queries with higher NDCG@10 under quantized embeddings, including English translations

Query	Topic (with English Translation)	Float	Quant	Δ
Q31	Boeteregeling / hoogte boete *(Penalty scheme / fine amount)*	0.0000	0.2891	+0.2891
Q132	Max. pagina's/slides Plan van Aanpak *(Maximum pages/slides Action Plan)*	0.4307	0.5000	+0.0693
Q5	Gunningscriteria / subgunningscriteria *(Award criteria / sub-award criteria)*	0.6502	0.6968	+0.0465
Q36	Overmachtsbepalingen (force majeure) *(Force majeure provisions)*	0.4306	0.4575	+0.0269
Q17	Geografisch werkgebied *(Geographical working area / scope)*	0.2184	0.2372	+0.0188
Q152	Gestanddoeningtermijn (dagen) *(Validity period / period of tender in days)*	0.4228	0.4416	+0.0188
Q124	Geschatte vs. maximale opdrachtwaarde *(Estimated vs. maximum contract value)*	0.7246	0.7366	+0.0120
Q140	Indexeringsregeling / prijsverhoging *(Indexation scheme / price increase terms)*	0.8072	0.8180	+0.0107
Q63	Bestandsindeling aanlevering documenten *(File format for document submission)*	0.9484	0.9582	+0.0098
Q72	Maximale opdrachtwaarde/plafondbudget *(Maximum contract value / ceiling budget)*	0.8374	0.8443	+0.0069
Q148	Overmachtsbepalingen (force majeure) *(Force majeure provisions)*	0.9505	0.9566	+0.0061
Q105	Facturatievereisten *(Invoicing requirements)*	0.6949	0.6995	+0.0045
Q95	Vormvereisten (taal, volgorde, bijlagen) *(Formal requirements: language, order, appendices)*	0.6803	0.6842	+0.0040

- **Q31 (Boeteregeling / hoogte boete [Penalty scheme / fine amount]):** This query achieved a dramatic improvement (+0.2891), climbing from a completely failing baseline of 0.0000 to 0.2891. Examination of the retrieval logs revealed that the Float model was severely distracted by tabular layout artifacts, ranking low-information text blocks containing currency symbols above the gold text. Int8 quantization effectively smoothed away these formatting noise distractions, allowing the core semantic concept of the penalty clause to surface. This qualitative log observation served as the primary rationale for engineering the explicit currency-injection configurations evaluated in the controlled synthetic test (see Section 3.7).

- **Procedural & Boilerplate Concepts (Q132 & Q5):** Themes like Plan van Aanpak formatting (Q132, +0.0693) and Gunningscriteria (Q5, +0.0465) benefit from looser semantic boundaries. Cross-referencing the query indices with the master dataset topic

categories reveals that standard administrative boilerplate clauses, such as force majeure frameworks (Q36, +0.0269 and Q148, +0.0061) or validity periods (Q152, +0.0188), demonstrated highly consistent, stable increments ranging from +0.0040 to +0.0269 under the quantized configuration.

4.2 English Retrieval Performance Evaluation

To assess the cross-lingual generalizability of the quantization effects, a parallel evaluation was conducted using an English translation of the procurement dataset. This configuration maintained an identical experimental setup: 172 queries evaluated against a candidate pool of 100 document snippets containing the respective gold-standard targets, comparing full-precision float embeddings with the `Int8`-quantized variant derived from the `multilingual-e5-large` model.

4.2.1 Aggregate Retrieval Quality

As documented in the English performance ledger provided in Figure 4.5, the aggregate performance of the English configuration reflects a small but noticeable improvement under quantization. The mean NDCG@10 score rises from 0.6059 for the float baseline to 0.6110 for the `Int8` variant ($\Delta = +0.0051$). A similar upward shift is observed in the Mean Reciprocal Rank (MRR), which increases from 0.6562 to 0.6601 ($\Delta = +0.0039$).

Metric	Float	Quantized
Mean NDCG@10	0.6059	0.6110
Mean MRR	0.6562	0.6601
Mean Kendall τ	0.9887 (pairwise)	

Figure 4.5: English Summary Scores: Comparison ledger of aggregate retrieval quality between full-precision baseline and Int8 scalar quantized configurations for the translated English control track.

At a first glance, these macro results demonstrate that the English translations behave in a remarkably uniform manner to the original Dutch dataset. Consistent with the primary Dutch findings reported in Section 4.1, these metrics confirm that the transition to an 8-bit embedding representation does not introduce systemic performance degradation at scale. Instead, the compression scheme seems to introduces minor, parallel structural shifts across both language tracks that marginally favor the quantized vector space.

4.2.2 Per-Query Variation and Stability

The expanded per-query distributions for NDCG@10 and MRR are illustrated vertically in Figure 4.6. Tracking the individual query indexes from top to bottom demonstrates that individual performance profiles remain exceptionally stable between the two separate vector configurations. Quantization preserves macro-level ranking integrity, an observation further validated by the micro-level tracking shown in the standalone rank correlation plot in Figure 4.7.

As with the previous Dutch evaluation track, these individual query charts isolate the first 50 query distributions purely to preserve line and bar legibility; mapping all 172 records simultaneously results in severe visual compression.

The pairwise rank correlation yields a high mean τ of 0.9887, tightly bounded within a narrow structural range (> 0.978). This close distribution confirms that despite minor isolated score fluctuations across different query indices, the relative permutation of candidate documents inside the 100-document pool remains highly consistent under matrix quantization constraints, closely mirroring the rank stability thresholds observed in the primary Dutch evaluation track.

4.2.3 Score Distribution and Quantization Artifacts

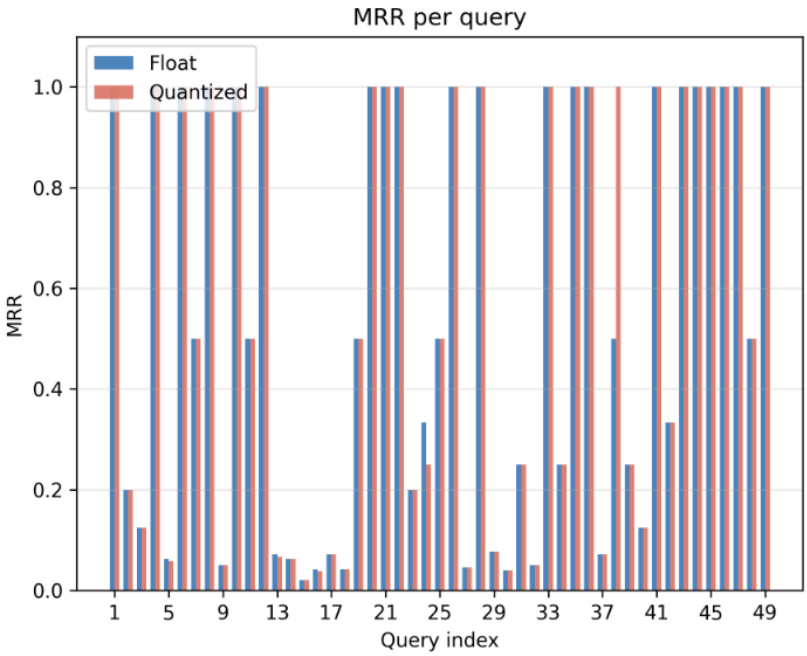
The characteristic cosine similarity inflation observed in the Dutch corpus is systematically replicated within the translated English control track. As illustrated by the frequency distributions in Figure 4.8, the full-precision float configuration yields gold document cosine similarities distributed around a mean of 0.8232 (median: 0.8237, range: $[0.7304, 0.8991]$). Upon applying `Int8` scalar quantization, this density distribution undergoes an aggressive rightward shift and structural compression, resulting in a significantly elevated mean similarity of 0.9631 (median: 0.9632, range: $[0.9438, 0.9790]$).

This localized crowding within the vector space is a direct manifestation of the uniform discretization artifacts outlined in Section 4.1.3. Because the automated translation workflow preserves the underlying contextual layout and semantic boundaries of the original procurement clauses, the resulting English embeddings concentrate within a highly restricted coordinate subspace. Forcing these dense continuous values into 256 discrete integer intervals flattens the subtle dimensional gradients that originally differentiated individual text snippets. Consequently, during vector re-normalization, the absolute similarity magnitudes are compressed into a narrow window approaching 1.0.

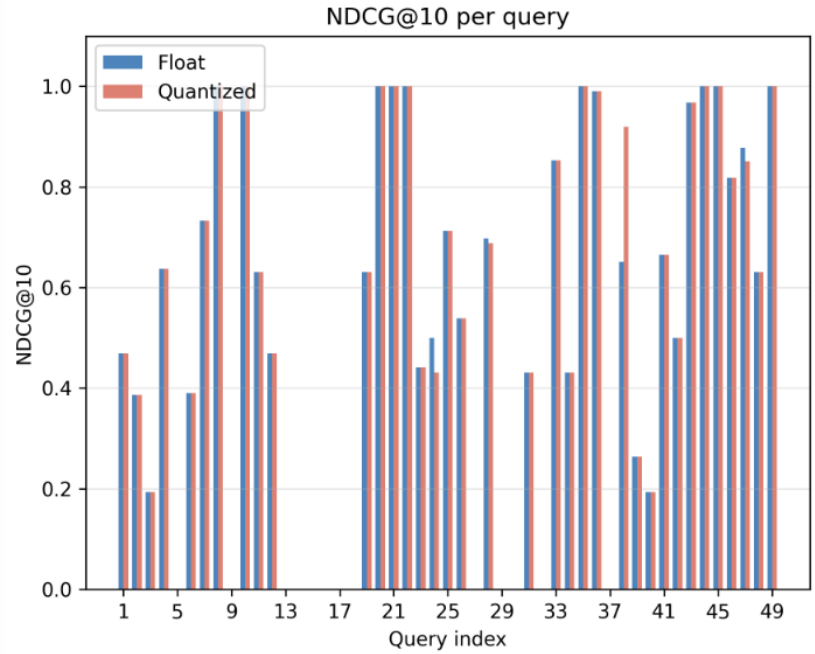
Crucially, downstream retrieval evaluations seem to confirm that this absolute score distortion remains entirely benign at the macro level. Consistent with the Dutch baseline analysis, ranking stability is maintained because the retrieval relies on the relative ordering of candidate vectors rather than their absolute geometric distances. While uniform integer mapping restricts the available coordinate resolution, it largely preserves the rank boundaries between segments. If a translated passage exhibits a strong semantic affinity for a given query in the high-resolution space, both candidates shift predictably rightward under `Int8` constraints, maintaining the target document's relative margin over adjacent competitors. The cross-lingual consistency of this crowding effect suggests that the phenomenon is primarily an architectural artifact of scalar quantization rather than a language-dependent variable.

4.2.4 Failure Case Analysis and Core Outlier Tracking

Just as observed within the Dutch processing track, a matched subset of the translated queries proved exceptionally challenging for both embedding configurations. While aggregate metrics remain robust at scale, a closer look at the localized errors reveals specific query indices where retrieval performance drops noticeably. An expanded cross-lingual comparative breakdown across language syntax boundaries will follow immediately in Section 4.3.



(a) Per-Query Reciprocal Rank (MRR)



(b) Per-Query Retrieval Quality (NDCG@10)

Figure 4.6: English Micro-Level Query Distributions: Expanded individual query index distributions (first 50 queries) for MRR and NDCG@10, illustrating the tightly synchronized performance overlap across the translated corpus.

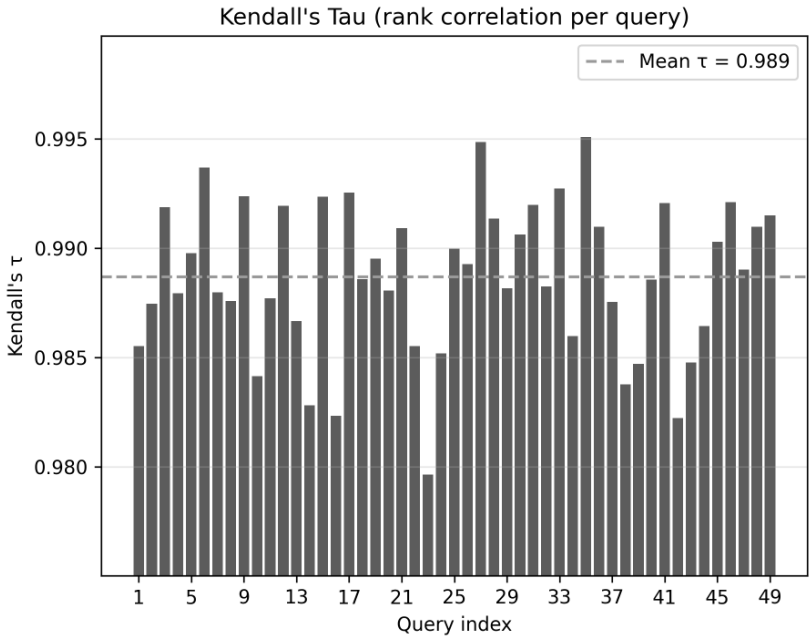
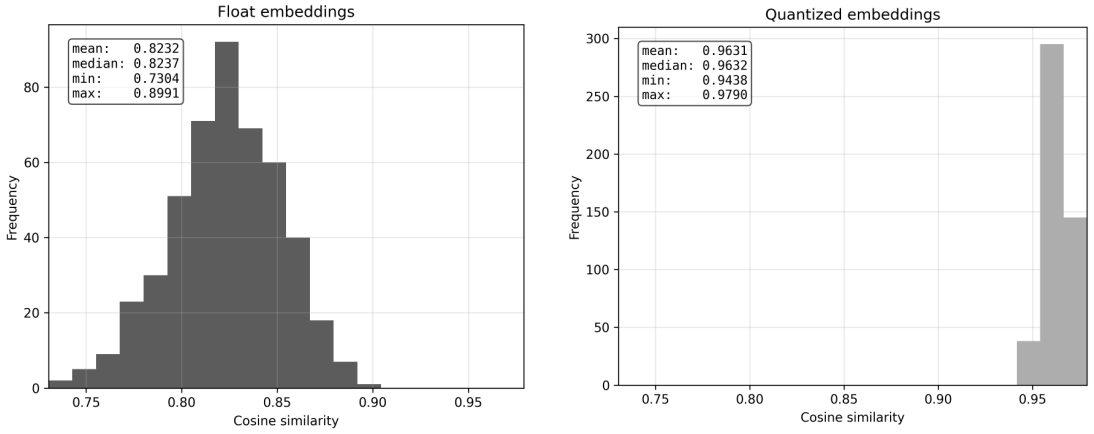


Figure 4.7: English Kendall's τ per Query: Individual rank correlation scores (for first 50 queries) mapping the stable structural consistency of document permutations under matrix compression.



(a) English Float Cosine Similarity

(b) English Quantized Cosine Similarity

Figure 4.8: English Geometric Similarity Scaling and Inflation: Frequency distributions comparing full-precision floating-point similarities against the Int8 compressed space, confirming an identical rightward density clustering profile under translation.

4.2.5 Micro-Level Degradations and Performance Gains

While the aggregate metrics established in Section 4.2.1 indicate global stability for the English track, a granular, query-by-query examination reveals distinct micro-level performance shifts when transitioning to low-precision integer spaces. This section breaks down individual translated English queries where the `Int8` scheme induced localized performance regressions (detailed in Table 4.5) or, conversely, triggered visible retrieval improvements due to semantic regularization (detailed in Table 4.6).

Analysis of Localized Regressions

Table 4.5: English queries with lower NDCG@10 under quantized embeddings

Query	Topic (English)	Float	Quant	Δ
Q135	Which ISO certifications are required	0.5000	0.4307	−0.0693
Q146	Waiting room construction specs	0.8772	0.8503	−0.0269
Q31	Penalty scheme fines for defaults	0.3562	0.3333	−0.0229
Q105	Invoicing requirements and conditions	0.6166	0.5991	−0.0175
Q139	Invoicing requirements (procedural)	0.6976	0.6886	−0.0089
Q117	Award criteria / scoring configurations	0.1070	0.1021	−0.0049
Q124	Estimated vs. Maximum order value	0.3638	0.3591	−0.0047

An analysis of the English regressions highlights important programmatic differences from the Dutch pipeline:

- **Vulnerability of Standardized Alphanumeric Identifiers (Q135):** The most severe micro-level degradation occurred in Q135 (ISO certification requirements), decreasing by −0.0693. Alphanumeric tokens and standardized system acronyms (such as "ISO 9001") rely on sharp dimensional representation vectors to execute exact lexical-semantic matching. Quantization to an 8-bit grid effectively compressed these high-variance sub-regions, preventing the compressed space from cleanly separating formal standard requirements from adjacent operational texts.

- **The Impact of Phrasal Expansion (Q31):** Counter to the Dutch track, where the compact target phrase (*Boeteregeling*) experienced a large performance boost under quantization, the translated English variant for Q31 (Penalty frameworks) suffered a minor regression of −0.0229. This cross-lingual divergence highlights a tokenization bottleneck. While Dutch legal terminology often relies on hyper-specific compound words, English translations expand these concepts into multi-word phrasal descriptions (e.g., "penalty scheme fines for defaults"). When these distributed textual phrases are compressed into an 8-bit integer grid, the boundary lines distinguishing the core semantic concept from adjacent boilerplate provisions probably got blurred, allowing irrelevant administrative text to marginally outrank the target gold snippet.

- **The Mitigation of Vector Outliers (Q107):** Crucially, within the English dataset, Q107 (Maximum budget ceilings) did not trigger the precision failure seen in the Dutch track (NDCG@10 = 0.3973). Reviewing the logs reveals that the machine translation pipeline translated the boilerplate text box containing the financial targets without

preserving the exact positioning of specialized structural artifacts (e.g., ⚠). Consequently, the scaling anomaly that skewed the Dutch scalar quantization boundaries was absent in the English embeddings, allowing the uniform linear intervals to partition cleanly.

Analysis of Localized Regularization Gains

Table 4.6: English queries with higher NDCG@10 under quantized embeddings

Query	Topic (English)	Float	Quant	Δ
Q150	Dispute settlements / Advisory Board	0.6509	0.9197	+0.2688
Q108	Consortium registration / Partnerships	0.0000	0.2891	+0.2891
Q40	Grace periods and contract extensions	0.0000	0.1772	+0.1772
Q104	Reference project submission parameters	0.6805	0.7933	+0.1128
Q77	Liability types and damage thresholds	0.4307	0.5000	+0.0693
Q50	Corporate Social Responsibility (CSR)	0.4367	0.4982	+0.0614
Q129	Geographical scope and location boundaries	0.3155	0.3333	+0.0179
Q83	Assessment methodology (procedural)	0.3854	0.3973	+0.0120
Q48	Award criteria / scoring configurations	0.4574	0.4694	+0.0120
Q18	Formal documentation and formatting	0.8194	0.8278	+0.0084
Q4	Assessment methodology and formulas	0.8626	0.8662	+0.0036
Q33	Subcontracting and third-party terms	0.8284	0.8320	+0.0036

The regularizing characteristics of Int8 scalar mapping are evident in these structural evaluations:

- **Substantial Baseline Jumps (Q108 & Q40):** Quantization acted as an outstanding baseline primer for Q108 (Partnerships) and Q40 (Grace periods), driving both up from absolute baseline failure (0.0000) to metrics of 0.2891 and 0.1772 respectively. A plausible theoretical explanation for this jump is that full-precision models frequently overfit onto local variations within generalized legal prose; compressing the semantic space down to 256 discrete levels may effectively operate as a low-pass filter. This compressed global variance is hypothesized to remove high-frequency lexical fluff, forcing nearby documents to merge into unified semantic clusters.
- **Resolution of Complex Administrative Concepts (Q150):** Q150 (Dispute settlements via the Advisory Board) achieved a substantial increase of +0.2688. Broad qualitative frameworks, such as arbitration guidelines, are easily distorted in float32 space by adjacent clauses sharing secondary procedural language (e.g., court jurisdictions, legal liability). The performance surge under the Int8 grid arrangement suggests that these continuous variances were compressed, potentially neutralizing local vocabulary differences and prioritizing the macro-level context of institutional dispute resolution mechanisms.

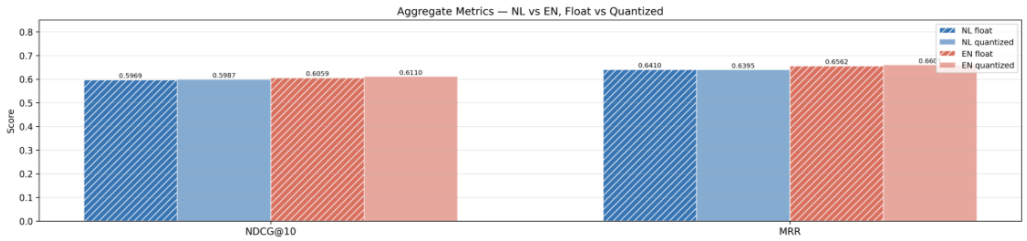
4.3 Cross-Lingual Comparative Analysis

Having established that both the original Dutch baseline and the translated English control track exhibit resilient macro-level performance independently, a cross-lingual comparative

analysis is needed to inspect the quantization behavior. While evaluating aggregate metrics in isolation suggests that the two language tracks behave very similar under compression, a side-by-side micro-level investigation reveals whether the observed stability is uniform across language representations. Comparing these parallel systems allows for a direct evaluation of how linguistic structure might alter the underlying geometry of compressed vector spaces.

4.3.1 Consistency of the Macro Shifts

At the macro level, aggregate performance trajectories behave similarly across both language tracks, verifying that the broad architectural limits of the model are stable under compression. As formalized in the comparative performance summary ledger in Figure 4.9, the English configuration consistently exhibits a marginal advantage over the original Dutch baseline across both precision states. The English model yields a mean NDCG@10 of 0.6059 in full precision and 0.6110 under quantization, whereas the Dutch configuration captures scores of 0.5969 and 0.5987.



(a) Aggregate Performance Metrics Overlap

Metric	NL	EN	Δ
NDCG@10 float	0.5969	0.6059	+0.0090
NDCG@10 quant	0.5987	0.6110	+0.0123
MRR float	0.6410	0.6562	+0.0152
MRR quant	0.6395	0.6601	+0.0206
Kendall τ	0.9882	0.9887	+0.0004
Mean score flt	0.8406	0.8232	-0.0174
n queries	172		
n gold docs	478		

(b) Comprehensive Performance Summary Ledger

Figure 4.9: Macro-Level Cross-Lingual Comparison: Distribution ledger and overlap bars showing identical aggregate upward trajectories for both Dutch and English datasets when transitioning to quantized indexes.

Both language profiles demonstrate a small, parallel upward shift in mean retrieval performance when switching to the compressed index space ($\Delta = +0.0018$ for Dutch and +0.0051 for English). This shows that at an aggregate level, the transition to an 8-bit embedding representation preserves macro-level retrieval quality in both evaluation tracks, suggesting that semantic structures survive the quantization process.

4.3.2 Asynchronous Micro-Level Volatility (Kendall's τ)

While aggregate changes follow nearly identical trends, a fine-grained exploration of individual query sequences reveals distinct, asynchronous fluctuations. As plotted in the expanded tracking lines in Figure 4.10 (truncated to the first 50 query indices to ensure graphic readability and maintain representativeness), both language tracks achieve a high baseline of rank preservation. However, tracking the line profiles reveals that the downward drops toward the 0.980 threshold occur on different query indices for each language configuration.

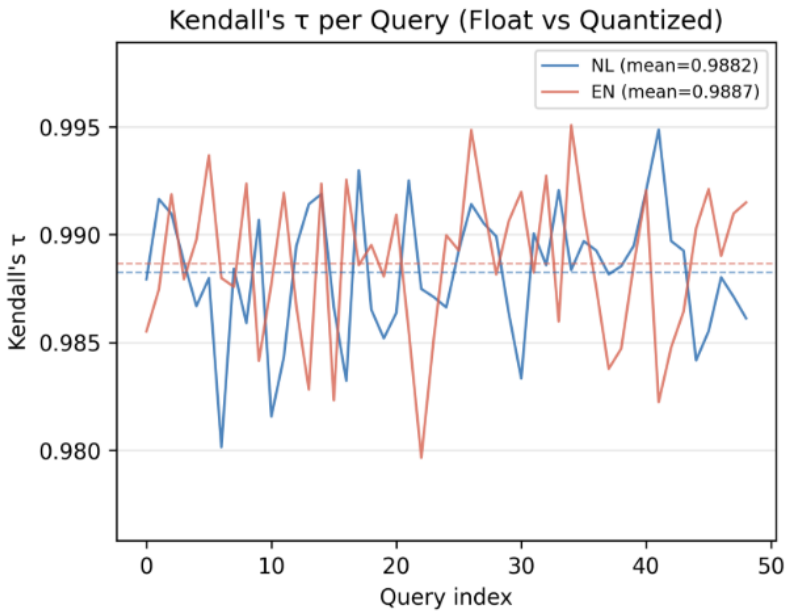


Figure 4.10: Asynchronous Rank Correlation Tracking: Comparative per-query Kendall's τ profiles (for first 50 queries), demonstrating a complete lack of horizontal synchronization in localized rank drops between the two language tracks.

A reduction in rank correlation for a specific Dutch query index does not match a corresponding drop in the English tracking line at that same query cross-section, and the reverse holds true as well. This complete lack of synchronization across the query horizontal axis shows that individual document permutations introduced by `Int8` rounding noise manifest independently across the Dutch and English vector representations. While the evaluation pipeline remains structurally identical, a major driver of this divergence is localized variation introduced during translation. Specifically, error analysis reveals that unique typographic symbols present in the Dutch corpus, which severely disrupted the Dutch quantized rounding thresholds were entirely absent from the English translations. Consequently, the rounding thresholds impact localized clusters differently because the same underlying concept is mapped to entirely unique coordinates and token distributions.

4.3.3 Pre-Quantization Distribution and Translative Pre-Smoothing

Rather than re-stating isolated statistics from the previous sections, examining the initial, unquantized vector distributions reveals why the English control track behaves differently from the original Dutch data. Comparing these baseline attributes isolates the impact of

the machine translation pipeline itself. As illustrated by the query delta distribution in Figure 4.11, the English variant produces higher overall metric scores across a majority of the outlying query positions.

This macro score advantage corresponds with geometric differences in the underlying vector distributions shown in Figure 4.12. Prior to quantization, the full-precision English model operates within a tighter, lower cosine similarity distribution window ($\mu = 0.8232$) compared to the Dutch unquantized subspace ($\mu = 0.8406$).

This tighter distribution indicates a lower standard deviation across the English embeddings; the data points cluster more uniformly together because the automated translation pipeline inherently acts as a pre-smoothing filter. Machine translation algorithms naturally tend to select standardized, common dictionary terms over the highly specific, regional legal-procurement idioms found in Dutch tenders. When the `Int8` mapping step is applied, both configurations shift rightward into an inflated, narrow window near 1.0. However, because the unquantized English subspace was already pre-smoothed and stripped of linguistic variance, it exhibits less sensitivity to downstream integer rounding boundaries, explaining the performance gains and asynchronous rank drops analyzed qualitatively in Chapter 5.

4.4 Controlled Validation via Synthetic Injection

To verify the underlying causes of the regressions observed across the Dutch and English pathways for the emoji sign in Q107, a controlled synthetic stress test was executed as described in 3.7. By holding the semantic context static while systematically altering structural variables, this experiment separated linguistic meaning from layout or formatting artifacts.

4.4.1 Empirical Results and Metric Analysis

The empirical findings demonstrate that inserting non-textual symbols, such as emojis or currency signs, disrupts the rounding of values during the Int8 compression stage. Under full precision (Float32), the model functions like a sharp lens, successfully distinguishing tiny, fine-grained differences between sentences. As illustrated in Panel A of Figure 4.13, the model maintains a amount of space between separate concepts in its internal map, allowing sentence pairs to spread out naturally along the continuum based on their unique semantic properties.

However, when these maps are compressed into the strict, 8-bit whole-number system (Int8), this fine resolution is completely flattened. As summarized in the experimental results, the sentences lose their distinct flavors. Instead of varying naturally, their similarity scores relative to the query are artificially inflated and packed into a narrow coordinate space near the upper margin.

This collapse is explicitly mapped out in Panel B of Figure 4.13. When a non-standard formatting element or emoji acts as an outlier token, it stretches the minimum and maximum boundaries of the coordinate system. Because uniform scale quantization relies on a rigid, evenly spaced grid, stretching the upper bound causes the individual integer bins to expand as well. Consequently, the standard text sentences (Pairs A, B, C, and D) are aggressively compressed and forced into the exact same narrow cluster of bins near the left margin.

This mechanism is comparable to taking a highly detailed grayscale image and increasing the contrast until all intermediate gray shades are crushed into pure black or white.

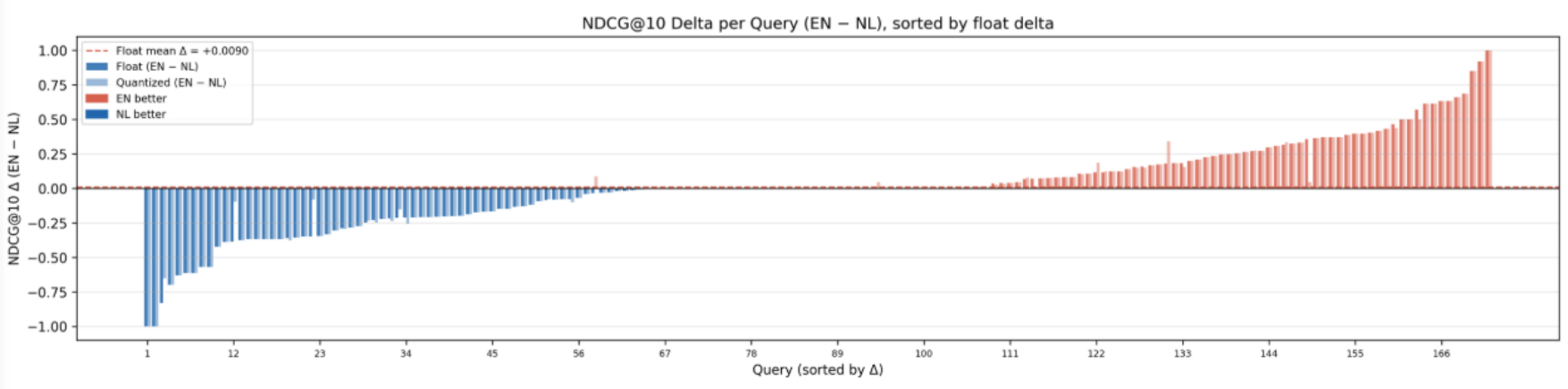


Figure 4.11: Linguistic Delta Overview: Per-query NDCG@10 variance showing that automated translation leads to an overall score advantage across a majority of outlying query indices.

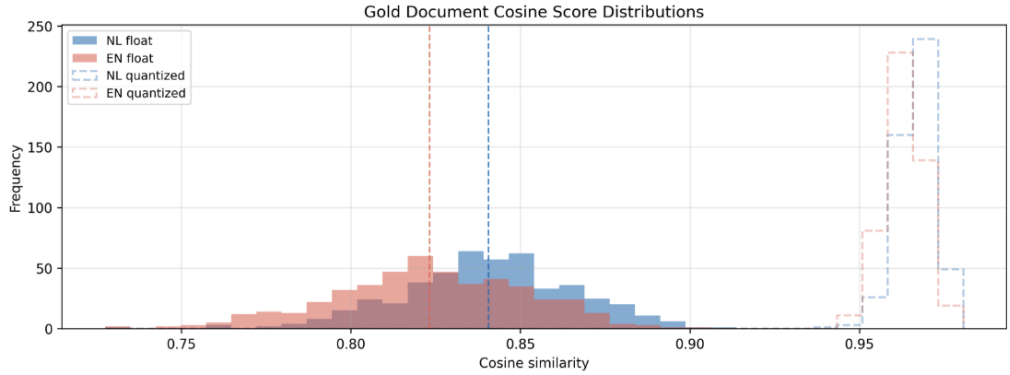


Figure 4.12: Cross-Lingual Score Space Overlap: Baseline and quantized frequency curves demonstrating the tighter pre-quantization variance of the English data and the uniform rightward shift under compression.

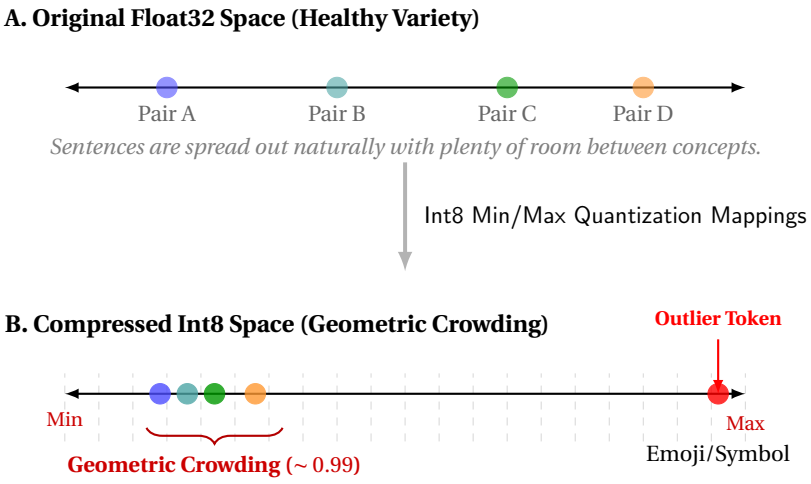


Figure 4.13: Empirically verified mechanism of axis-stretching and geometric crowding. The introduction of an outlier character stretches the calibration scale, packing standard text vectors into identical integer boxes.

Because the text items are compressed together, they cluster tightly around an unnatural, near-perfect similarity score of ~ 0.99 relative to the query. Since these separate sentences look identical to the retrieval engine, the model loses the ability to determine the most accurate match. This text-scrambling behavior reduces the ranking correlation down to a significantly degraded Kendall's τ score of 0.8112 ($p = 1.9794 \times 10^{-16}$) compared to the Float32 baseline. This statistical variance matches the performance drops found in the log tracking logs, showing that the retrieval pipeline encounters severe localized limits when faced with specialized layout characters.

This blurring mechanism is explicitly visualized via the pairwise similarity heatmaps in Figure 4.14. The Float32 matrix displays clear, structured differentiation between disparate sentences. In contrast, the Int8 matrix collapses into a nearly uniform, solid green block. This visually demonstrates the absolute loss of fine-grained resolution when separate text snippets are compressed into identical integer brackets.

4.4.2 The Quantization Mechanism: Axis-Stretching and Bin Mismatch

To find the exact mathematical trigger behind this performance drop, the coordinate shifts across individual embedding dimensions were tracked. Figure 4.15 displays the percentage of dimensions that shifted into different integer boxes when a single formatting symbol or currency prefix was added to an otherwise identical text string.

An investigation was conducted to determine whether these symbols were actually changing the meaning of the sentences or simply breaking the calculations during compression. When looking at the original, high-precision vectors (Float32), the change in meaning was almost zero ($\Delta\cos \leq 0.0342$ across all tests, and as low as 0.0131 for pair P19). Because the actual meaning barely changed, it became clear that the issue was happening at the coordinate level when the numbers were rounded. Remarkably, this tiny tweak triggers a massive coordinate reorganization under quantization. An average of 90.60% of all 1,024 vector dimensions are forced to jump into a completely wrong integer box, peaking at a maximum mismatch rate of 93.26% for the currency-modified text block in pair P18.

This massive grid mismatch occurs because standard 8-bit compression relies on a rigid, evenly spaced grid. This grid is scaled entirely by the absolute minimum and maximum values found across a vector axis. Because text models treat non-standard symbols and currency markers as extreme mathematical outliers, these specialized characters artificially stretch the grid boundaries.

Consequently, the rounding scale becomes way too wide. This forces the remaining 95% of normal text data to get squeezed into the exact same small handful of integer boxes. With 90.60% of the dimensions forced into these distorted boxes, the fine details that separate different sentence meanings are completely overwritten by structural noise.

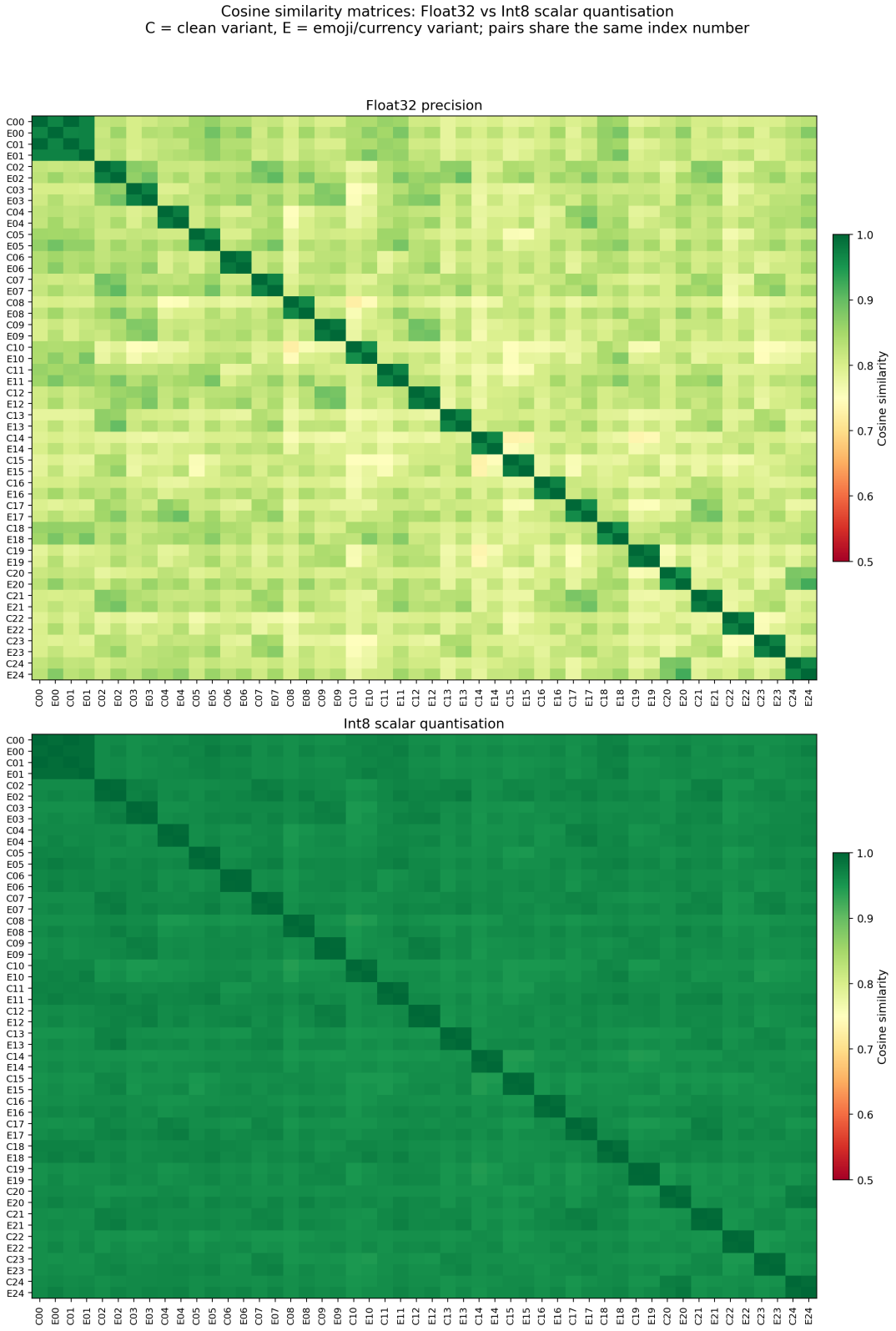


Figure 4.14: Pairwise cosine similarity matrices demonstrating the visual collapse from healthy geometric variance in Float32 (top) to a uniform, blurred similarity space under Int8 Quantization (bottom).

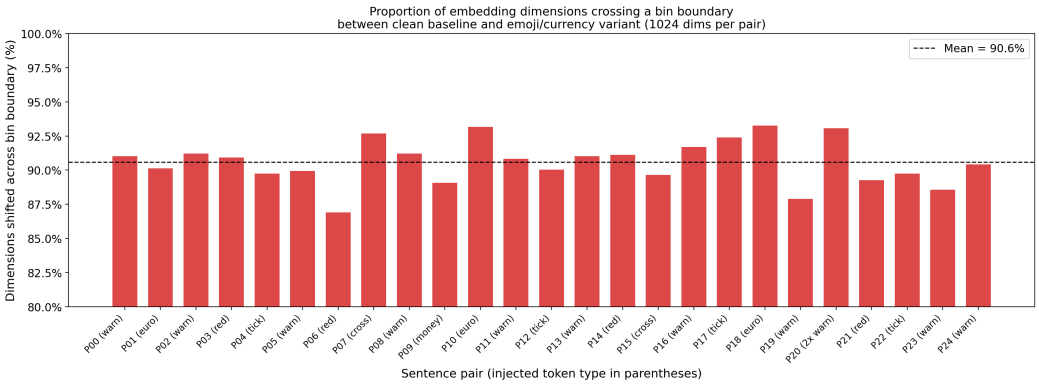


Figure 4.15: Dimension-level impact showing the percentage of vector bins flipped by the insertion of structural tokens across all 25 pairs for Int8 precision.

Chapter 5

Discussion

The empirical findings presented in Chapter 4 show that while aggregate macro-metrics remain remarkably stable under Int8 quantization, this resilience masks a highly volatile micro-level landscape. Transitioning from Float32 into Int8 dimensions forces an shift from broad statistical averages to localized semantic variations. This chapter contextualizes these shifts, moving beyond raw empirical tracking to explore the underlying geometric, linguistic, and architectural mechanisms that govern Int8 quantization.

5.1 Dataset-Wide (Macro) Geometry: The Paint-by-Numbers Collapse

To understand why search results shift after compression, it helps to look at how the mathematical boundaries of the grid affect the data as a whole.

5.1.1 Quantization as a Semantic Filter

An unexpected benefit of compressing data into 8-bit integers (Int8) is that it can actually improve simple, broad search queries. In full-precision space, text embeddings often capture unnecessary details like punctuation choice, writing style, or standard formatting that do not matter for matching core concepts.

In theory, compression is intended to flatten minor, low-information variations into identical grid coordinates. This filtering is designed to help a search engine focus on the core meaning of the text rather than superficial phrasing choices. However, this smoothing effect relies heavily on the assumption that the discarded detail is purely noise. In practice, this assumption risks failing entirely when processing highly specific technical terms, specialized data, or formatting elements that distort the underlying grid scale.

5.1.2 The Compression Grid and Geometric Crowding

Standard 8-bit quantization projects a vast, multi-dimensional floating-point space onto a rigid grid bounded strictly between -128 and 127. This conversion alters how different text concepts are separated, driving the dramatic upward shift in similarity scores seen across the results.

This systematic drift is best explained by a “Paint-by-Numbers” effect. In full precision, the model acts like an artist mixing custom, fluid oil paints to capture infinite shades of meaning. Forcing these precise decimal values onto a rigid, 256-level integer grid strips

away these subtle shades. Because specialized procurement datasets naturally share a narrow vocabulary, the uniform grid leaves the vast majority of its available integer boxes completely empty. The system is essentially forced to recreate a complex landscape using only a fraction of its available color palette.

When these simplified representations are scaled to calculate search scores, the loss of detail causes distinct documents to crowd tightly together near a similarity score of 1.0. This dataset-wide crowding alters the entire search landscape, squeezing the empty space between unrelated concepts and making the system highly sensitive to minor calculation changes.

5.2 Word-Level (Micro) Linguistics: How Tokenization Shapes Rounding

The fact that search rankings drop in one language but remain stable in another proves that compression errors do not affect all text equally. Instead, errors depend heavily on how a specific language's word structure is mapped onto the grid boxes.

5.2.1 Subword Fragmentation vs. Whole Word Phrasing

The primary cause of this language divide is how words are broken down by multilingual models. Because a shared vocabulary must accommodate over 100 languages, smaller or more grammatically complex languages are broken down far more aggressively into fragments.

This creates an imbalance in how concepts look geometrically:

- **The English Layout (Distributed Whole Words):** In English, concepts usually appear as chains of separate, high-frequency words (e.g., *"notice period conditions"*). Because these words exist as complete, native entries in the vocabulary, their mathematical weight is spread smoothly and evenly across multiple tokens. This keeps the vector values low and safely in the center of the grid boxes, protecting English search paths from rounding errors.

- **The Dutch Layout (Aggressive Subword Fragmentation):** Conversely, Dutch packs identical concepts into long, complex compound nouns (e.g., *"opzeggingstermijn"*). Because these compounds are missing from the shared vocabulary, the system chops them into abstract subword pieces (such as `_gestand`, `doen`, `ing`, `term`, `ijn`). To reconstruct the unified concept, the model must compress these fragments into sharp, high-amplitude spikes on specific coordinate dimensions.

As visualized in Figure 5.1, these sharp, localized spikes are highly vulnerable to rounding boundaries. Because English phrases split their meaning smoothly across multiple common words, their vector values sit safely in the stable center of the integer grid boxes. Conversely, a Dutch compound noun forces the model to bundle complex meaning into high-amplitude spikes. Because uniform quantization scales the grid based on the maximum values in the vector, these peaks are mathematically pushed to the absolute edges of the rounding intervals. This amplification shifts the values dangerously close to the dividing lines between integer steps, where even a tiny floating-point variation forces the coordinate to snap into a completely different bin.

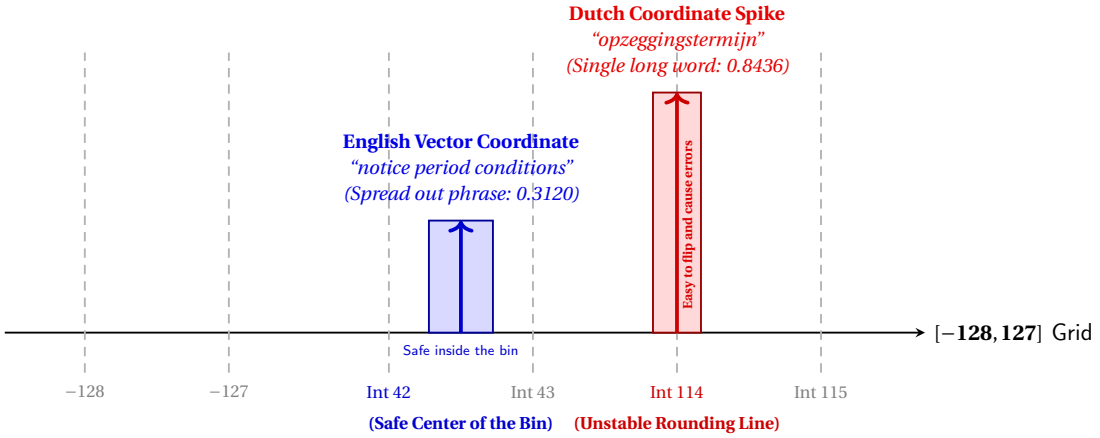


Figure 5.1: How word structure affects rounding: A visual comparison showing how long compound words (Dutch) form tall coordinate spikes that sit dangerously on rounding lines, while separate words (English) form lower values that sit safely in the center of integer bins.

As a result, even small changes in the data can cause these high Dutch coordinates to tip over into an adjacent box. This localized boundary-crossing alters the vector's final orientation, creating a subtle drift in meaning that causes the mismatched search rankings shown in Figure 4.10.

5.3 Outlier Tokens and Axis-Stretching Theory

The vulnerability of uniform linear quantization becomes acute when text strings contain non-standard structural elements, such as currency signs, legal formatting marks, or emojis. While these tokens carry little to no semantic meaning, their presence mathematically disrupts the entire quantization grid.

When a document or query contains an outlier character, the model generates extreme coordinate spikes along specific hidden dimensions to represent this distinct visual symbol. Because a uniform quantization grid is forced to span the absolute maximum and minimum values across an entire axis, a single extreme outlier forces the boundaries of that grid to widen drastically.

This expansion causes a severe loss of resolution for the rest of the text. Because the grid scale stretches to accommodate the outlier, the remaining coordinates that are representing the actual text content are then compressed into a tiny fraction of the available integer boxes.

This structural distortion explains why similarity matrices blur into uniform blocks and why an overwhelming majority of vector dimensions are forced to snap into incorrect bins. Subtle meaning differences are completely erased as standard text sequences are aggressively crowded into identical integer brackets. Ultimately, this mechanism strips the retrieval engine of its capacity for high-precision differentiation, proving that layout characters can completely overwrite semantic boundaries with structural noise.

5.4 Architectural Implications for High-Precision Industry Pipelines

These insights carry profound implications for the design of industrial information retrieval and legal tech systems, particularly for software vendors operating in localized, high-resource languages.

5.4.1 The Deception of Macro-Benchmarks

The most critical takeaway for production system architects is that macro-evaluation metrics, such as aggregate mean NDCG@10 or MRR are deceptive. In high-stakes industries like procurement automation (e.g., Brainial) or legal informatics, system failure does not occur across the average of thousands of queries; it occurs when a single, high-value document or specific legal condition is completely missed due to localized rank corruption.

Relying on broad macro-benchmarks creates a false sense of security, masking severe micro-level regressions (such as the localized failures analyzed in Section 4.1.5 and Section 4.2.5) that directly undermine user trust and operational reliability.

5.4.2 Designing for Production Resilience

To mitigate these vulnerabilities, production pipelines utilizing Int8 constraints must move away from naive, uniform scalar quantization. Because industry documents are inherently noisy containing mixed formatting, tables, currency symbols, and complex compound vocabulary, architectures should look into outlier-preserving or block-wise quantization frameworks.

Furthermore, when operating in highly morphological languages like Dutch, embedding models must be paired with customized tokenizers that avoid aggressive subword fragmentation, ensuring that domain-specific technical terminology is mapped smoothly across the latent space rather than forced into volatile geometric spikes.

Chapter 6

Conclusion

6.1 Summary of Empirical Findings and Key Takeaways

This thesis presented a systematic, query-level exploration into the impact of 8-bit (Int8) scalar embedding quantization on high-precision information extraction within the highly specialized domain of Dutch procurement. By evaluating the `e5-multilingual-large` model across parallel Dutch and machine-translated English pipelines, this research balanced engineering efficiency against semantic fidelity. The empirical investigations yielded four primary insights:

- **Dataset-Wide (Macro) Metric Resilience:** At an aggregate level, transitioning from full-precision (Float32) to compressed (Int8) configurations resulted in negligible performance changes. On the original Dutch dataset, the mean NDCG@10 scores exhibited a minute baseline increase from 0.5969 to 0.5987, while the Mean Reciprocal Rank (MRR) shifted from 0.6410 to 0.6395. This global resilience proves that 8-bit scalar quantization achieves its core objective of a 75% reduction in the index memory footprint and enables accelerated integer arithmetic without undermining dataset-wide retrieval capabilities.

- **The Micro-Level Paradox (Regularization vs. Distortion):** Granular error tracking exposed localized, query-level fluctuations that were completely masked by macro averages. Counterintuitively, compression frequently functioned as a beneficial semantic filter. By smoothing away continuous stylistic clutter and unnecessary phrasing noise, it forced nearby documents into unified semantic clusters. This elevation lifted failing baseline queries up to functional benchmarks, such as Q31 (which contained heavy table formatting) jumping by +0.2891. Conversely, localized performance drops were tied directly to exact alphanumeric criteria (such as ISO certifications), which act as mathematical outliers that warp quantization intervals.

- **The Catastrophic Impact of Structural Outliers:** Controlled testing using a specialized synthetic dataset revealed that non-textual structural elements, such as emojis and currency signs, trigger a distinct failure mode defined as axis-stretching. Because a uniform quantization grid must span from the absolute minimum to the maximum values found along a coordinate axis, a single extreme outlier character forces the calibration boundaries to widen drastically. This expansion packs standard text coordinates into a tiny handful of identical integer boxes near the margin, causing a severe loss of resolution where distinct sentences cluster tightly around a blurred similarity

score of ~ 0.99. This geometric crowding eliminates the model's ability to rank documents accurately, reducing the ranking correlation down to a weak Kendall's τ score of 0.8112.

- **Asynchronous Cross-Lingual Behavior:** The parallel cross-lingual baseline analysis showed that micro-level query volatility manifests independently across language tracks. The complete lack of synchronization across languages highlighted that integer rounding noise does not degrade a universal, language-agnostic concept. Instead, localized ranking shifts depend heavily on how a specific language's structural syntax maps a topic into the vector space. Compact compound words inherent to administrative Dutch (e.g., *gunningscriteria*, translated as *"award criteria"*) create high-amplitude coordinate spikes that sit dangerously on rounding thresholds, making them highly vulnerable to bin-flipping errors. Meanwhile, distributed, multi-token English equivalents buffer the system against extreme localized document shuffling.

Ultimately, this study demonstrates that while Int8 embedding quantization is an exceptionally viable engineering mechanism for sustainable, production-scale retrieval pipelines, its deployment in high-stakes legal-administrative domains demands a cautious approach. Automated procurement pipelines cannot evaluate safety based on macro benchmarks alone. Systems must actively account for localized microscopic vulnerabilities induced by formatting anomalies and language-specific geometric drifts to guarantee uniform, high-precision retrieval reliability.

6.2 Future Research

While this study demonstrates that Int8 embedding quantization maintains remarkable macro-level stability for information extraction in specialized domains, the localized anomalies observed in the micro-level error analysis open distinct avenues for future investigation:

1. **Mitigation Strategies for Axis-Stretching and Outliers:** The synthetic stress tests executed in this study suggested that non-textual tokens, such as emojis and currency signs, trigger destructive axis-stretching. However, the exact threshold at which an outlier character begins to degrade the grid remains unmapped. Future work could investigate whether varying the density or position of these characters alters the severity of the collapse.

   Additionally, researchers should evaluate alternative quantization frameworks to fix this vulnerability. Testing methods that weren't covered in this research like outlier-aware quantization could possibly protect the retrieval engine from geometric crowding without sacrificing the memory benefits of 8-bit storage.

2. **Native Multilingual Corpora vs. Translation Smoothing:** The cross-lingual analysis revealed that automated machine translation pipelines inherently regularize text, smoothing out obscure regional procurement jargon and artificially inflating baseline metrics before embeddings are even generated.

   To isolate the true impact of language-specific syntax under quantization, future studies should benchmark the Dutch pipeline against a manually curated, original English procurement corpus containing verified industry-specific jargon. This approach would bypass the pre-smoothing bias introduced by automated translation workflows.

Expanding this boundary to investigate how quantization impacts low-resource languages or highly localized legal dialects within shared embedding spaces would further clarify how compression constraints impact specific language families.

3. **Morphological Complexity and Vocabulary Fragmentation:** This study highlighted how the aggressive fragmentation of long Dutch compound words (such as *gunningsleidraad*, meaning "bidding guidelines") creates high-amplitude vector spikes that are highly vulnerable to rounding errors.

   Future research could systematically isolate this morphological variable by comparing how quantization treats intact compound terms against variants that are explicitly pre-segmented with hyphens or spaces. Testing whether deliberate compound splitting before the embedding stage stabilizes retrieval rankings would help design text preprocessing pipelines tailored specifically for compressed multilingual models.

Appendix A - Data statement

A. Curation Rationale

The dataset was curated to evaluate the precision-efficiency trade-offs of embedding quantization in a high-stakes, specialized domain. Specifically, it aims to test how well a compressed vector space preserves technical, legal, and administrative jargon unique to Dutch procurement. The corpus consists of five distinct, publicly available Dutch tenders split into 16,819 individual text snippets ("boxes"). To evaluate retrieval performance, 172 "Points of Attention" (POAs) were established as research queries.

B. Language Variety

- **BCP-47 Tag:** `nl-NL` (Mainstream Dutch as used in the Netherlands).
- **Prose Description:** Technical, administrative, and legal Dutch jargon typical of procurement and official government tender documents in the Netherlands.

C. Speaker Demographic

- **Age:** Adult professionals.
- **Gender:** Mixed.
- **Race/Ethnicity:** Unknown.
- **Native Language:** Dutch (`nl-NL`) assumed.
- **Socioeconomic Status:** Professional/Corporate class.
- **Number of Speakers:** Multiple corporate and governmental authors representing the creators of the 5 distinct tenders.
- **Presence of Disordered Speech:** None.

D. Annotator Demographic

The ground-truth labels ("Gold Standard") were established by a single annotator.

- **Age:** Between 35–40
- **Gender:** Male

- **Race/Ethnicity:** Dutch
- **Native Language:** Dutch
- **Socioeconomic Status:** Professional class.
- **Training/Expertise:** Professional Senior Tender Manager and sole domain specialist available at Brainial B.V.
- **Inter-Annotator Agreement (IAA):** N/A (*A recognized limitation of this dataset is the absence of a second annotator to validate the labels or provide a reliability metric*).

E. Speech Situation

- **Time and Place:** Publicly available tenders in the Netherlands, gathered up to 2026.
- **Modality:** Written text.
- **Scripted/Edited vs. Spontaneous:** Highly edited, formal, professional corporate/governmental text.
- **Synchronous vs. Asynchronous Interaction:** Asynchronous (static official documentation).
- **Intended Audience:** Commercial vendors, contractors, data analysts, and procurement platforms.

F. Text Characteristics

The corpus consists of 16,819 text snippets across 169 unique documents with an average length of 17.4 words per snippet (totaling 293,309 words). The text is dominated by legal constraints, technical building requirements, contract terms, penalty clauses, and administrative acronyms.

G. Recording Quality

N/A (Strictly textual data).

H. Other

Curator Demographics: The dataset compilation and text segmentation were handled in collaboration with Brainial B.V. The supervision on the thesis project was done under academic supervision at Vrije Universiteit Amsterdam.

I. Provenance Appendix

N/A (Built directly from primary public tender documentation provided by Brainial B.V.).

Appendix B - Synthetic Emoji Stress Test Raw Logs

This appendix provides the full empirical log output from the controlled synthetic stress-test described in Section 3.7. The experiment tracks 50 English sentences structured as 25 pairs ($P00$ through $P24$). For each pair, an alphanumeric control sentence is evaluated against its structurally modified variant containing layout symbols or financial notations. All items are measured against the baseline target query: *"The maximum contract value for this project is capped at 500000..."*.

.1 Sentence Pairs

In Table 1, the raw sentence pairs are displayed. Note that within the Python implementation, the actual Unicode emoji characters were passed directly to the model; text placeholders (e.g., `[Warning Icon]`) are utilized in the table below solely for PDF rendering and cross-platform compilation compatibility.

Table 1: Full qualitative configuration inventory detailing the 25 manually engineered evaluation sentence pairs.

Pair	Control (Alphanumeric Baseline)	Treatment (Injected Semantic Twin)
P00	The maximum contract value for this project is capped at 500000	[Warning Icon] The maximum contract value for this project is capped at 500000
P01	The maximum contract value for this project is capped at 500000	The maximum contract value for this project is capped at €500000
P02	All bids must be submitted before the deadline in the tender documents	[Warning Icon] All bids must be submitted before the deadline in the tender documents
P03	The supplier must deliver all goods within 30 days of contract award	[Red Circle] The supplier must deliver all goods within 30 days of contract award
P04	Eligible candidates must have at least five years of relevant experience	[Checkmark] Eligible candidates must have at least five years of relevant experience
P05	The contractor shall provide proof of insurance with minimum coverage of 1000000	[Warning Icon] The contractor shall provide proof of insurance with minimum coverage of 1000000
P06	The project completion deadline is set for December 31 2026	[Red Circle] The project completion deadline is set for December 31 2026
P07	Subcontractors must be approved in writing before work commences	[Cross Mark] Subcontractors must be approved in writing before work commences
P08	The tendering authority reserves the right to cancel this procurement at any time	[Warning Icon] The tendering authority reserves the right to cancel this procurement at any time

Table 1: Full qualitative configuration inventory detailing the 25 manually engineered evaluation sentence pairs.

Pair	Control (Alphanumeric Baseline)	Treatment (Injected Semantic Twin)
P09	Payment will be made within 60 days of receipt of a valid invoice	[Cash Bag] Payment will be made within 60 days of receipt of a valid invoice
P10	The estimated budget for phase two of the project is 750000	[Cash Bag] The estimated budget for phase two of the project is €750000
P11	Tenderers must submit a financial guarantee equal to 5 percent of the contract value	[Warning Icon] Tenderers must submit a financial guarantee equal to 5 percent of the contract value
P12	The winning bidder will be notified within 10 working days after evaluation	[Checkmark] The winning bidder will be notified within 10 working days after evaluation
P13	All documentation must be submitted in both printed and electronic format	[Warning Icon] All documentation must be submitted in both printed and electronic format
P14	The scope of work includes design, construction, and commissioning of the facility	[Red Circle] The scope of work includes design, construction, and commissioning of the facility
P15	Late submissions will not be considered under any circumstances	[Cross Mark] Late submissions will not be considered under any circumstances
P16	The contract is subject to the laws and jurisdiction of the Netherlands	[Warning Icon] The contract is subject to the laws and jurisdiction of the Netherlands
P17	Candidates must demonstrate experience in public procurement procedures	[Checkmark] Candidates must demonstrate experience in public procurement procedures
P18	The minimum turnover requirement for this tender is 2000000 per year	[Cash Bag] The minimum turnover requirement for this tender is €2000000 per year
P19	Liquidated damages apply at a rate of 0.1 percent per day of delay	[Warning Icon] Liquidated damages apply at a rate of 0.1 percent per day of delay
P20	The authority may extend the contract by up to two additional years	[Dual Warning Icons] The authority may extend the contract by up to two additional years
P21	Tenderers must be registered in the national procurement database	[Red Circle] Tenderers must be registered in the national procurement database
P22	The evaluation criteria include price quality and sustainability	[Checkmark] The evaluation criteria include price quality and sustainability
P23	Security clearance is required for all personnel working on-site	[Warning Icon] Security clearance is required for all personnel working on-site
P24	The total contract duration including optional extensions shall not exceed 4 years	[Warning Icon] The total contract duration including optional extensions shall not exceed 4 years

.2 Aggregate Statistical Summary

Before looking at individual sentence trajectories, the macro properties across the entire 50-sentence synthetic tracking space reveal a severe global compression and rank degradation under Int8 linear scalar quantization configurations:

- **Ranking Correlation Decay:** Kendall's $\tau = 0.8112 \quad (p = 1.9794 \times 10^{-16})$
- **Mean Coordinate Bin Mismatch Rate:** 90.60% of all vector dimensions

- **Maximum Boundary Displacement Rate:** 93.26% (Observed in Pair $P18$)
- **Minimum Boundary Displacement Rate:** 86.91% (Observed in Pair $P06$)
- **Outlier Vector Coordinate Detections:** None (0.00%) — Every embedding coordinate fell within standard calibration limits, confirming that performance drops stem entirely from internal scaling factor compression rather than hard boundary clipping.

.3 Detailed Sentence Pair Tracking Records

Table 2 lists the precise mathematical displacements across all 25 sentence configurations, detailing the structural modification token type, the baseline full-precision similarity, the collapsed Int8 similarity, the absolute semantic drift (Δcos), and the resulting coordinate integer box flipping rate (Bins Δ%).

Table 2: Granular geometric tracking records of the 25 evaluation sentence pairs under Float32 precision vs. Int8 scalar quantization mappings.

Pair	**Token Type**	**Float cos**	**Quant cos**	**Δ cos**	**Bins Δ%**
P00	Warning	0.9690	0.9934	−0.0245	91.0%
P01	Currency	0.9718	0.9941	−0.0223	90.1%
P02	Warning	0.9746	0.9947	−0.0200	91.2%
P03	Circle	0.9745	0.9946	−0.0201	90.9%
P04	Checkmark	0.9773	0.9952	−0.0179	89.7%
P05	Warning	0.9707	0.9939	−0.0232	89.9%
P06	Circle	0.9828	0.9964	−0.0136	86.9%
P07	Cross	0.9713	0.9939	−0.0226	92.7%
P08	Warning	0.9676	0.9932	−0.0256	91.2%
P09	Currency	0.9824	0.9963	−0.0139	89.1%
P10	Currency	0.9593	0.9914	−0.0321	93.2%
P11	Warning	0.9712	0.9940	−0.0228	90.8%
P12	Checkmark	0.9802	0.9959	−0.0157	90.0%
P13	Warning	0.9708	0.9939	−0.0231	91.0%
P14	Circle	0.9674	0.9932	−0.0258	91.1%
P15	Cross	0.9780	0.9954	−0.0174	89.6%
P16	Warning	0.9718	0.9941	−0.0223	91.7%
P17	Checkmark	0.9623	0.9921	−0.0298	92.4%
P18	Currency	0.9595	0.9915	−0.0320	93.3%
P19	Warning	0.9833	0.9964	−0.0131	87.9%
P20	Warning	0.9567	0.9909	−0.0342	93.1%
P21	Circle	0.9799	0.9958	−0.0158	89.3%
P22	Checkmark	0.9741	0.9946	−0.0205	89.7%
P23	Warning	0.9741	0.9946	−0.0205	88.6%
P24	Warning	0.9691	0.9935	−0.0245	90.4%

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